Week 9: Logit and Probit

Samuel Rowe adapted from Wooldridge (2011), Long and Freese (2006), and Mitchell (2020)

2025-10-29

Contents

```r
bookdown::render_book()
```

Overview

We will cover the implementation of logit, probit, and multinomial logit.

1. Logit, Probit, and Linear Probability Model
2. Multinomial Logit
3. Ordinal Logit
4. Reshape

Chapter 1

Binary Outcomes

Married Women's Labor Force Participation

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use mroz.dta, clear
```

We'll use the data from Mroz (1987) to look at the probability of a married woman being in the labor force. Labor force participation is a binary response.

$$y = [0, 1]$$

We will estimate the coefficients of the linear probability model (LPM), the logit estimator, and the probit estimator. Then, we'll compare the marginal effects ofall three estimators.

Summarize in the labor force

```
tabulate inlf
```

```
       inlf |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        325       43.16       43.16
          1 |        428       56.84      100.00
------------+-----------------------------------
      Total |        753      100.00
```

There are 325 women are not in the labor force and 428 women participating in the labor force. Our explanatory variables are non-wife income, education, experience, experience-squared, age, kids less than 6, kids greater than 6

$$y_i = \beta_0 + \beta_1 spouseinc_i + \beta_2 edu_i + \beta_3 exp_i + \beta_4 exp_i^2 + \beta_5 kidsLT6_i + \beta_6 kidsGT6_i + \varepsilon_i$$

```
est clear
eststo Logit: logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
eststo Probit: probit inlf nwifeinc educ exper expersq kidslt6 kidsge6
esttab Logit Probit, mtitle
```

```
Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -422.78042
Iteration 2:   log likelihood = -421.73851
Iteration 3:   log likelihood = -421.73502
Iteration 4:   log likelihood = -421.73502

Logistic regression                             Number of obs     =        753
                                                LR chi2(6)        =     186.28
                                                Prob > chi2       =     0.0000
Log likelihood = -421.73502                     Pseudo R2         =     0.1809

------------------------------------------------------------------------------
        inlf |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |  -.0301171   .0082431    -3.65   0.000    -.0462734   -.0139609
        educ |   .2520038   .0425492     5.92   0.000      .168609    .3353987
       exper |   .2057387   .0310518     6.63   0.000     .1448784     .266599
     expersq |   -.003913   .0009994    -3.92   0.000    -.0058718   -.0019541
     kidslt6 |  -.9175126   .1742458    -5.27   0.000    -1.259028   -.5759971
     kidsge6 |   .2226164   .0683456     3.26   0.001     .0886616    .3565713
       _cons |  -3.739707    .543217    -6.88   0.000    -4.804392   -2.675021
------------------------------------------------------------------------------


Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -422.36847
Iteration 2:   log likelihood = -421.80202
Iteration 3:   log likelihood = -421.80161
Iteration 4:   log likelihood = -421.80161

Probit regression                               Number of obs     =        753
                                                LR chi2(6)        =     186.14
                                                Prob > chi2       =     0.0000
Log likelihood = -421.80161                     Pseudo R2         =     0.1808

------------------------------------------------------------------------------
```

inlf	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
nwifeinc	-.017188	.00474	-3.63	0.000	-.0264782	-.0078978
educ	.1501412	.02471	6.08	0.000	.1017105	.1985719
exper	.1240105	.0183233	6.77	0.000	.0880975	.1599236
expersq	-.0023694	.0005913	-4.01	0.000	-.0035284	-.0012103
kidslt6	-.5543317	.1038244	-5.34	0.000	-.7578238	-.3508395
kidsge6	.1307901	.0399186	3.28	0.001	.0525511	.2090292
_cons	-2.244553	.3146254	-7.13	0.000	-2.861207	-1.627899

	(1) Logit	(2) Probit
inlf		
nwifeinc	-0.0301***	-0.0172***
	(-3.65)	(-3.63)
educ	0.252***	0.150***
	(5.92)	(6.08)
exper	0.206***	0.124***
	(6.63)	(6.77)
expersq	-0.00391***	-0.00237***
	(-3.92)	(-4.01)
kidslt6	-0.918***	-0.554***
	(-5.27)	(-5.34)
kidsge6	0.223**	0.131**
	(3.26)	(3.28)
_cons	-3.740***	-2.245***
	(-6.88)	(-7.13)
N	753	753

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

We cannot compare the coefficients across the models. We will need to use marginal effects.

Average Marginal Effects (AME)

We will first look at average marginal effects. For average marginal effects, we estimate the marginal effects for each i and estimate an average.

$$AME = (\sum_{i=1}^{n}[g(\hat{\beta}_0 + x\hat{\beta})\beta_j]\Delta x_j)/n$$

Linear Probability Model (LPM)

Please note that we need the option *post* to start marginal effects.

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo LPM: margins, dydx(*) post
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Logit: margins, dydx(*) post
```

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Probit: margins, dydx(*) post
```

Compare

```
esttab LPM Logit Probit, mtitle
```

```
                      (1)             (2)             (3)   
                      LPM           Logit          Probit   
------------------------------------------------------------
nwifeinc         -0.00341*       -0.00381*       -0.00362*  
                  (-2.35)         (-2.57)         (-2.51)   

educ               0.0380***       0.0395***       0.0394***
                   (5.15)          (5.41)          (5.45)   
```

```
exper               0.0395***       0.0368***       0.0371***
                    (6.96)          (7.14)          (7.20)

expersq          -0.000596**     -0.000563**     -0.000568**
                   (-3.23)         (-3.18)         (-3.20)

age                -0.0161***      -0.0157***      -0.0159***
                   (-6.48)         (-6.60)         (-6.74)

kidslt6             -0.262***       -0.258***       -0.261***
                   (-7.81)         (-8.07)         (-8.20)

kidsge6             0.0130          0.0107          0.0108
                    (0.99)          (0.81)          (0.83)
------------------------------------------------------------
N                      753             753             753
------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

Marginal Effects at the Average (MEA)

For the marginal effects at the average, we set our x to their means within the scalar $g(.)$

$$MEA = g(\hat{\beta}_0 + \hat{\beta}_1 \bar{x_1} + ... + \hat{\beta}_k \bar{x_k})\beta_j \Delta x_j$$

Linear Probability Model LPM

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo LPM: margins, dydx(*) atmeans post
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Logit: margins, dydx(*) atmeans post
```

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo Probit: margins, dydx(*) atmeans post
```

Compare

```
esttab LPM Logit Probit, mtitle
```

```
                      (1)             (2)             (3)
                      LPM           Logit          Probit
------------------------------------------------------------
nwifeinc         -0.00341*       -0.00519*       -0.00470*
                  (-2.35)         (-2.53)         (-2.48)

educ               0.0380***       0.0538***       0.0511***
                   (5.15)          (5.09)          (5.19)

exper              0.0395***       0.0501***       0.0482***
                   (6.96)          (6.40)          (6.57)

expersq         -0.000596**     -0.000767**     -0.000737**
                  (-3.23)         (-3.10)         (-3.14)

age               -0.0161***      -0.0214***      -0.0206***
                  (-6.48)         (-6.05)         (-6.24)

kidslt6            -0.262***       -0.351***       -0.339***
                  (-7.81)         (-7.07)         (-7.32)

kidsge6            0.0130          0.0146          0.0141
                   (0.99)          (0.80)          (0.83)
------------------------------------------------------------
N                     753             753             753
------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The analysis shows that the marginal effects are fairly close across the linear probability model, Logit model, and Probit model. One additional year of education increases the probability of being in the labor force by a range of 0.038 to 0.0395 percentage points. Interestingly, one additional child less than six is associated with a drop in the probability of being in the labor force by a range of 0.258 to 0.262.

Please not that around the means, our linear probability model, Logit, and Probit should be fairly similar. However, the marginal effects for the linear probability model are constant and will not vary across different values of x.

Odds Ratios

We can use the option, or to get odds ratios after running a logit.

$$OR = \frac{(OddsSuccess)}{(OddsFailure)} = \frac{p(1)/(1-p(1))}{p(0)/(1-p(0))}$$

```
logit inlf nwifeinc educ exper expersq age kidslt6 kidsge6, or
```

```
Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -402.38502
Iteration 2:   log likelihood = -401.76569
Iteration 3:   log likelihood = -401.76515
Iteration 4:   log likelihood = -401.76515

Logistic regression                             Number of obs     =        753
                                                LR chi2(7)        =     226.22
                                                Prob > chi2       =     0.0000
Log likelihood = -401.76515                     Pseudo R2         =     0.2197

------------------------------------------------------------------------------
        inlf | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |    .978881   .0082436    -2.53   0.011     .9628565    .9951723
        educ |   1.247536   .0541925     5.09   0.000     1.145717    1.358404
       exper |   1.228593   .0393849     6.42   0.000     1.153775    1.308263
     expersq |   .9968509   .0010129    -3.10   0.002     .9948676    .9988381
         age |   .9157386   .0133451    -6.04   0.000     .8899527    .9422715
     kidslt6 |   .2361344   .0480734    -7.09   0.000      .158441    .3519257
     kidsge6 |   1.061956   .0794234     0.80   0.422     .9171603     1.22961
       _cons |   1.530283   1.316609     0.49   0.621     .2834155    8.262655
------------------------------------------------------------------------------
```

One additional year of education is associated with a 1.25 times increase in the odds of being in the labor force (or an increase of 25%) holding all other variables constant. One additional child less than six decreases the odds of being in the labor force by a factor of 0.24 holding all other variables constant (or a decrease of 76%).

Marginal Effects of Education at different points along the curve

Linear Probability Model (LPM)

```
est clear
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo lpm: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

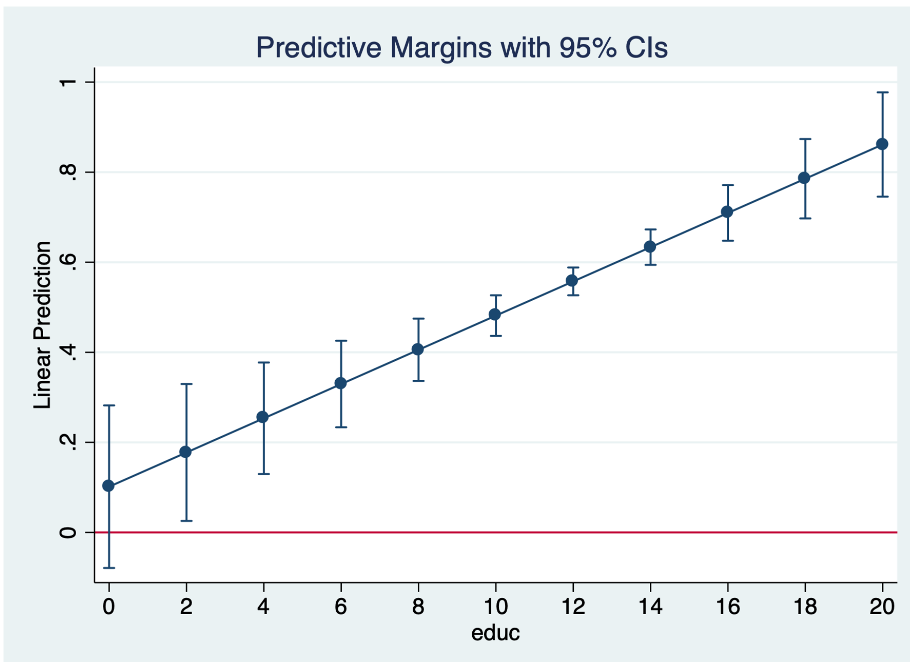

Figure 1.1: Marginal Effect of Education

Logit

```
quietly logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
eststo logit1: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

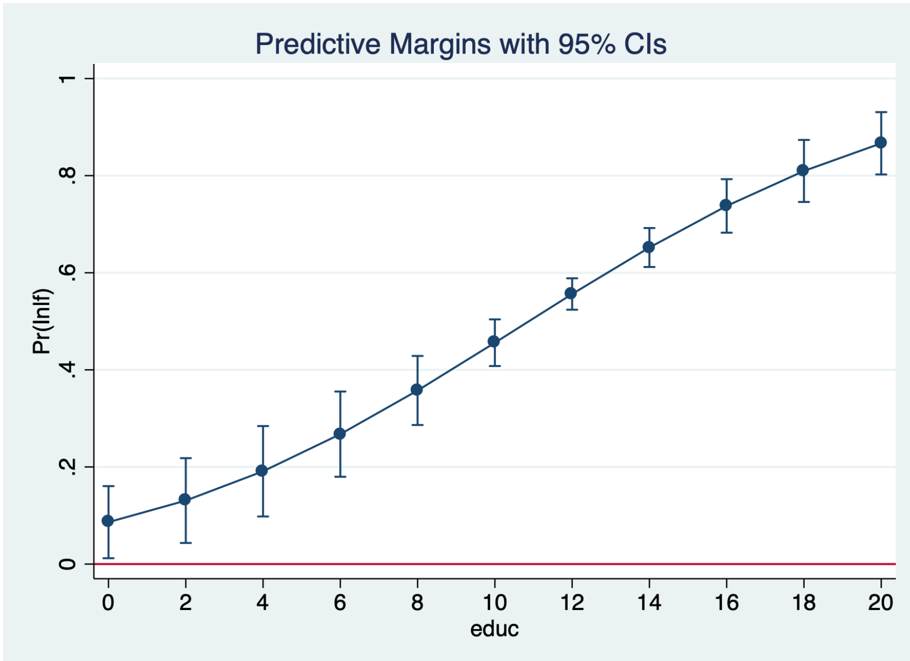

Figure 1.2: Marginal Effect of Education

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
eststo probit1: margins, at(educ=(0(2)20)) post
marginsplot, yline(0)
```

The predicted probability that a married women is in the labor force rises from 47.7% for 12 years of education to 71.2% for 16 years of education.

Coefficient Plot

We can also use coefplot to plot our data.

```
coefplot lpm logit1, at recast(line) ciopts(recast(rline) lpattern(dash))
coefplot lpm probit1, at recast(line) ciopts(recast(rline) lpattern(dash))
coefplot logit1 probit1, at recast(line) ciopts(recast(rline) lpattern(dash))
```

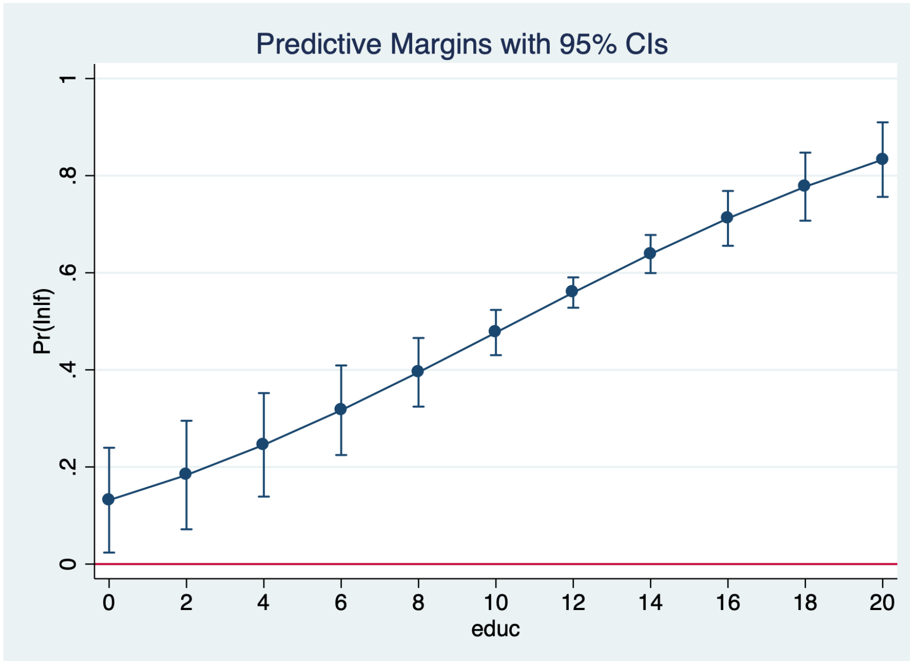

Figure 1.3: Marginal Effect of Education

Marginal Effects of Education at different points along the curve

Linear Probability Model (LPM)

```
quietly reg inlf nwifeinc educ exper expersq age kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
```

Logit

```
quietly logit inlf nwifeinc educ exper expersq kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_logitinlf.png", replace
```

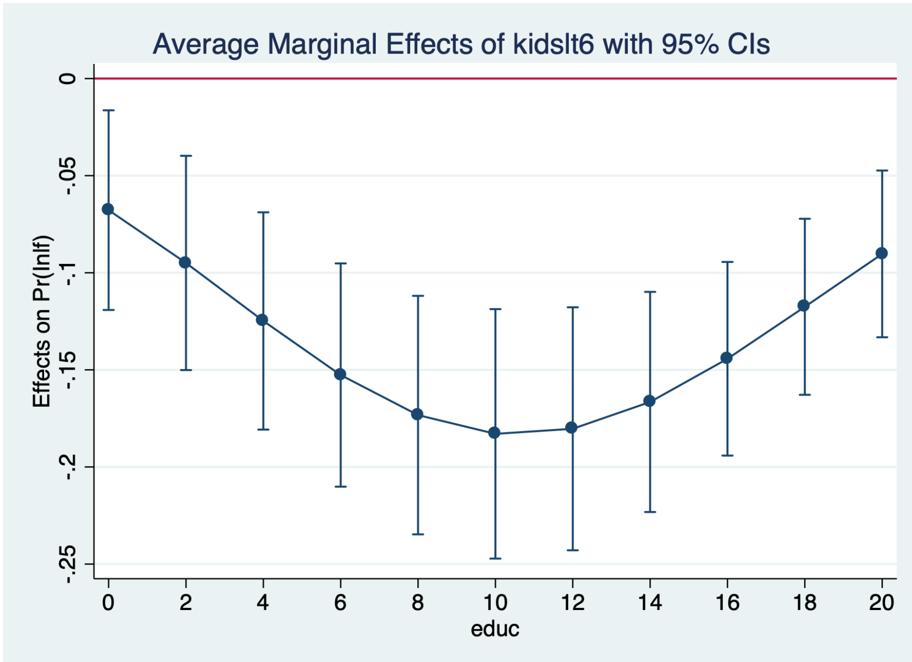

Figure 1.4: Marginal Effect of Kids Less Than 6

Probit

```
quietly probit inlf nwifeinc educ exper expersq age kidslt6 kidsge6
margins, dydx(kidslt6) at(educ=(0(2)20))
marginsplot, yline(0)
```

The average marginal effect for an additional child less than 6 rises from -18.3 percentage points to -14.4 percentage points, but the difference does not appear to be statistically significant.

Chapter 2

Multinomial Logit

Adapted from Long and Freese (2006)

When we have nominal categorical variables, we cannot use a binary logit or probit. We will need to use multinomial logit or probit.

Lesson: Interpreting a nominal categorical dependent variable We will use Current Population Survey Data from September 2023 and 2024 to estimate the following model for labor force participation:

$$lfs_i = \beta_0 + \beta_1 edu_i + \beta_2 exper_i + ... + u_i$$

.

There are three alternatives: Employed, Unemployed, and Not in the Labor Force.

Example Current Population Survey

```
use "/Users/Sam/Desktop/Econ 645/Data/CPS/mlogit_example.dta", clear
tab laborforce
tab laborforce, nolabel
```

```
Labor Force |
     Status |      Freq.     Percent        Cum.
------------+-----------------------------------
   Employed |     23,760       57.12       57.12
 Unemployed |        832        2.00       59.12
       NILF |     17,007       40.88      100.00
```

```
------------+-----------------------------------
      Total |     41,599      100.00
```

```
Labor Force |
     Status |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |     23,760       57.12       57.12
          2 |        832        2.00       59.12
          3 |     17,007       40.88      100.00
------------+-----------------------------------
      Total |     41,599      100.00
```

Our dependent variable has three nominal categories.

$$y = [1, 2, 3]$$

r

$$y = [Employed, Unemployed, NILF]$$

We use the *mlogit* command to run a multinomial logit to get log odds for $J-1$ logits.

```
mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.mar
```

```
Iteration 0:   log likelihood = -31774.039
Iteration 1:   log likelihood = -22975.759
Iteration 2:   log likelihood = -22620.368
Iteration 3:   log likelihood = -22573.244
Iteration 4:   log likelihood = -22565.789
Iteration 5:   log likelihood = -22564.343
Iteration 6:   log likelihood = -22564.014
Iteration 7:   log likelihood = -22563.941
Iteration 8:   log likelihood = -22563.925
Iteration 9:   log likelihood = -22563.922
Iteration 10:  log likelihood = -22563.922
Iteration 11:  log likelihood = -22563.922
Iteration 12:  log likelihood = -22563.922

Multinomial logistic regression                 Number of obs     =     41,599
                                                LR chi2(38)       =   18420.23
                                                Prob > chi2       =     0.0000
Log likelihood = -22563.922                     Pseudo R2         =     0.2899

------------------------------------------------------------------------------------
```

```
        laborforcestatus |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------------------+----------------------------------------------------------------
Employed                 |  (base outcome)
-------------------------+----------------------------------------------------------------
Unemployed               |
                  educat |
                     HSD |  -.2279524   .1119685    -2.04   0.042    -.4474066   -.0084982
            Some College |  -.4312492   .1303621    -3.31   0.001    -.6867542   -.1757442
           AA/Vocational |  -.6931682    .165487    -4.19   0.000    -1.017517   -.3688197
                   BS/BA |  -.6706784   .1331892    -5.04   0.000    -.9317245   -.4096323
Graduate or Professional |  -1.063383   .1745188    -6.09   0.000    -1.405433   -.7213323
                         |
                     exp |  -.0291051   .0088759    -3.28   0.001    -.0465016   -.0117087
                    exp2 |   .0002524   .0001575     1.60   0.109    -.0000562    .0005611
                         |
          race_ethnicity |
  Asian/Pacific Islander |   -.813795   .3012644    -2.70   0.007    -1.404262   -.2233275
                   Black |  -.3951124   .2756281    -1.43   0.152    -.9353337    .1451088
         Hispanic/Latino |  -.6504449   .2691511    -2.42   0.016    -1.177971   -.1229184
                   White |  -.7855686   .2616547    -3.00   0.003    -1.298402   -.2727347
             Multiracial |  -.4798449   .3366208    -1.43   0.154     -1.13961    .1799197
                         |
                  female |
                  Female |  -.0004137   .0721979    -0.01   0.995     -.141919    .1410915
                         |
               metroarea |
           Nonmetro Area |  -.1010417    .097797    -1.03   0.302    -.2927204    .0906369
          Not Identified |   .0635435    .346152     0.18   0.854     -.614902     .741989
                         |
                   union |
                   Union |  -19.66515   2339.277    -0.01   0.993    -4604.564    4565.234
                         |
                 marital |
    Divorced/Sep/Widowed |   .6132556   .1175463     5.22   0.000     .3828691     .843642
           Never Married |   .7526873   .0988405     7.62   0.000     .5589634    .9464112
                         |
                 hryear4 |
                    2024 |   .0228376    .071126     0.32   0.748    -.1165667    .1622419
                         |
                   _cons |  -2.023219   .3012462    -6.72   0.000    -2.613651   -1.432787
-------------------------+----------------------------------------------------------------
NILF                     |
                  educat |
                     HSD |  -.7898216   .0425444   -18.56   0.000    -.8732071   -.7064361
            Some College |  -.8941042   .0479568   -18.64   0.000    -.9880978   -.8001105
           AA/Vocational |  -1.143176   .0559087   -20.45   0.000    -1.252755   -1.033597
```

```
                   BS/BA  |  -1.493236   .0488335   -30.58   0.000    -1.588948   -1.39
Graduate or Professional  |  -1.746904   .0561914   -31.09   0.000    -1.857038   -1.63
                          |
                      exp |  -.1490983    .003087   -48.30   0.000    -.1551487    -.14
                     exp2 |   .0031412   .0000469    66.92   0.000     .0030492    .003
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.1995846   .1336975    -1.49   0.135    -.4616269    .062
                   Black  |  -.1053158   .1293382    -0.81   0.415     -.358814    .148
        Hispanic/Latino   |  -.4785722   .1270357    -3.77   0.000    -.7275575   -.229
                   White  |  -.3378929   .1237786    -2.73   0.006    -.5804944   -.095
             Multiracial  |  -.1802636   .1537778    -1.17   0.241    -.4816625    .121
                          |
                   female |
                  Female  |   .5781835   .0258942    22.33   0.000     .5274318    .628
                          |
               metroarea  |
           Nonmetro Area  |  -.0069684   .0328081    -0.21   0.832    -.0712712    .057
          Not Identified  |  -.0021328   .1282129    -0.02   0.987    -.2534254    .249
                          |
                    union |
                   Union  |  -20.25621   566.9973    -0.04   0.972     -1131.55    1091
                          |
                  marital |
     Divorced/Sep/Widowed |  -.0531382   .0373157    -1.42   0.154    -.1262757    .019
           Never Married  |   .1464745   .0379587     3.86   0.000     .0720769    .220
                          |
                  hryear4 |
                    2024  |  -.0398405   .0253899    -1.57   0.117    -.0896037    .009
                          |
                    _cons |   1.072448   .1365064     7.86   0.000     .8049003    1.33
-----------------------------------------------------------------------------------------
Note: 2044 observations completely determined.  Standard errors questionable.
```

Next, we need to test our Independence of Irrelevant Alternatives (IIA) assumption with a Hausman Test.

Estimate an Unrestricted model - We'll remove unemployed

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.uni
estimates store unrestricted
```

We compare the log odds for unemployed and not in the labor force to being employed. Given that these are log odds, we'll need to convert them to Odds Ratios or find the marginal effects.

Estimate a Restricted Model to test Independence of Irrelevant Alternatives assumption

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.marit
estimate store restricted
```

Use hausman command

```
hausman restricted unrestricted, alleqs constant
```

```
Note: the rank of the differenced variance matrix (2) does not equal the number of
        coefficients being tested (19); be sure this is what you expect, or there may be
        problems computing the test.  Examine the output of your estimators for anything
        unexpected and possibly consider scaling your variables so that the coefficients are
        on a similar scale.

                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |   restricted  unrestricted    Difference          S.E.
-------------+----------------------------------------------------------------
      educat |
          2  |   -.7860864    -.7898216        .0037352        .0035785
          3  |     -.89149    -.8941042        .0026142        .0032032
          4  |   -1.140012    -1.143176         .003164         .003669
          5  |   -1.486705    -1.493236        .0065306        .0036976
          6  |   -1.737118    -1.746904        .0097867        .0035815
         exp |   -.1486206    -.1490983        .0004777        .0002218
        exp2 |    .0031335     .0031412       -7.78e-06        2.91e-06
race_ethni~y |
          2  |   -.1712505    -.1995846        .0283341        .0120289
          3  |   -.0717454    -.1053158        .0335704        .0121412
          4  |   -.4427331    -.4785722        .0358391        .0115676
          5  |   -.3060009    -.3378929         .031892        .0117006
          6  |   -.1491388    -.1802636        .0311247        .0136923
    1.female |    .5793374     .5781835        .0011539        .0018992
   metroarea |
          2  |   -.0070074    -.0069684        -.000039         .002025
```

```
         3  |   -.0064057    -.0021328       -.0042729       .0112617
    1.union |   -20.20846    -20.25621        .0477553              .
    marital |
         2  |   -.0524486    -.0531382        .0006896       .0022895
         3  |    .1495841     .1464745        .0031096       .0023193
      _cons |     1.00979     1.072448       -.0626578              .
------------------------------------------------------------------------------
                          b = consistent under Ho and Ha; obtained from mlogit
           B = inconsistent under Ha, efficient under Ho; obtained from mlogit

    Test:  Ho:  difference in coefficients not systematic

                  chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =        8.92
                Prob>chi2 =      0.0116
                (V_b-V_B is not positive definite)
```

Our chi-squared is 8.92, which means with reject the IIA assumption. We should consider a binary response here.

Predicted Probabilities

Next, we use Stata to estimate predicted probabilities

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.un
margins, atmeans predict(outcome(1))
margins, atmeans predict(outcome(2))
margins, atmeans predict(outcome(3))
```

```
Adjusted predictions                            Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Employed), predict(outcome(1))
at           : 1.educat        =    .1261088 (mean)
               2.educat        =     .286954 (mean)
               3.educat        =    .1488497 (mean)
               4.educat        =    .0969254 (mean)
               5.educat        =    .2079617 (mean)
               6.educat        =    .1332003 (mean)
               exp             =    32.89247 (mean)
               exp2            =    1467.595 (mean)
               1.race_eth~y    =    .0098079 (mean)
               2.race_eth~y    =    .0622611 (mean)
               3.race_eth~y    =    .0931032 (mean)
```

```
                4.race_eth~y    =    .1487776 (mean)
                5.race_eth~y    =    .6685738 (mean)
                6.race_eth~y    =    .0174764 (mean)
                0.female        =    .4804923 (mean)
                1.female        =    .5195077 (mean)
                1.metroarea     =    .8000673 (mean)
                2.metroarea     =    .1904132 (mean)
                3.metroarea     =    .0095195 (mean)
                0.union         =    .9508642 (mean)
                1.union         =    .0491358 (mean)
                1.marital       =    .5163586 (mean)
                2.marital       =    .1819515 (mean)
                3.marital       =    .3016899 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       _cons |   .7684672   4.830264     0.16   0.874    -8.698677    10.23561
------------------------------------------------------------------------------


Adjusted predictions                            Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Unemployed), predict(outcome(2))
at           : 1.educat        =    .1261088 (mean)
               2.educat        =     .286954 (mean)
               3.educat        =    .1488497 (mean)
               4.educat        =    .0969254 (mean)
               5.educat        =    .2079617 (mean)
               6.educat        =    .1332003 (mean)
               exp             =    32.89247 (mean)
               exp2            =    1467.595 (mean)
               1.race_eth~y    =    .0098079 (mean)
               2.race_eth~y    =    .0622611 (mean)
               3.race_eth~y    =    .0931032 (mean)
               4.race_eth~y    =    .1487776 (mean)
               5.race_eth~y    =    .6685738 (mean)
               6.race_eth~y    =    .0174764 (mean)
               0.female        =    .4804923 (mean)
               1.female        =    .5195077 (mean)
               1.metroarea     =    .8000673 (mean)
               2.metroarea     =    .1904132 (mean)
               3.metroarea     =    .0095195 (mean)
               0.union         =    .9508642 (mean)
```

```
               1.union         =    .0491358 (mean)
               1.marital       =    .5163586 (mean)
               2.marital       =    .1819515 (mean)
               3.marital       =    .3016899 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       _cons |   .0090578    1.03352     0.01   0.993    -2.016605     2.03472
------------------------------------------------------------------------------


Adjusted predictions                            Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==NILF), predict(outcome(3))
at           : 1.educat        =    .1261088 (mean)
               2.educat        =     .286954 (mean)
               3.educat        =    .1488497 (mean)
               4.educat        =    .0969254 (mean)
               5.educat        =    .2079617 (mean)
               6.educat        =    .1332003 (mean)
               exp             =    32.89247 (mean)
               exp2            =    1467.595 (mean)
               1.race_eth~y    =    .0098079 (mean)
               2.race_eth~y    =    .0622611 (mean)
               3.race_eth~y    =    .0931032 (mean)
               4.race_eth~y    =    .1487776 (mean)
               5.race_eth~y    =    .6685738 (mean)
               6.race_eth~y    =    .0174764 (mean)
               0.female        =    .4804923 (mean)
               1.female        =    .5195077 (mean)
               1.metroarea     =    .8000673 (mean)
               2.metroarea     =    .1904132 (mean)
               3.metroarea     =    .0095195 (mean)
               0.union         =    .9508642 (mean)
               1.union         =    .0491358 (mean)
               1.marital       =    .5163586 (mean)
               2.marital       =    .1819515 (mean)
               3.marital       =    .3016899 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
```

```
       _cons |   .2224751   4.825216     0.05   0.963    -9.234774    9.679725
------------------------------------------------------------------------------
```

Average Marginal Effects

```
margins, dydx(*) predict(outcome(1))
margins, dydx(*) predict(outcome(2))
margins, dydx(*) predict(outcome(3))
```

```
Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female
               2.metroarea 3.metroarea 1.union 2.marital 3.marital

--------------------------------------------------------------------------------------------
                           |            Delta-method
                           |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
---------------------------+----------------------------------------------------------------
                    educat |
                      HSD  |   .1309012   .0072675    18.01   0.000     .1166572    .1451452
             Some College  |   .1499942   .0080928    18.53   0.000     .1341326    .1658558
            AA/Vocational  |   .1911979   .0091537    20.89   0.000     .1732569    .2091389
                    BS/BA  |   .2407492   .0078942    30.50   0.000     .2252768    .2562215
Graduate or Professional   |   .2789487    .008527    32.71   0.000     .2622362    .2956613
                           |
                       exp |   .0219808   .0004288    51.26   0.000     .0211403    .0228213
                      exp2 |  -.0004585   5.91e-06   -77.62   0.000      -.00047   -.0004469
                           |
            race_ethnicity |
  Asian/Pacific Islander   |   .0417403   .0212777     1.96   0.050     .0000368    .0834438
                    Black  |   .0224391   .0206352     1.09   0.277    -.0180052    .0628834
          Hispanic/Latino  |   .0802357   .0202145     3.97   0.000     .0406159    .1198555
                    White  |   .0618266   .0197636     3.13   0.002     .0230907    .1005625
              Multiracial  |   .0348149    .024377     1.43   0.153    -.0129632     .082593
                           |
                    female |
                   Female  |  -.0839096   .0039186   -21.41   0.000    -.0915899   -.0762292
                           |
                 metroarea |
             Nonmetro Area |   .0022776   .0050425     0.45   0.652    -.0076056    .0121607
```

```
         Not Identified  |  -.0007629   .0197304    -0.04   0.969    -.0394337    .03
                         |
                   union |
                   Union |   .4414329   .0020464   215.71   0.000     .4374219    .44
                         |
                 marital |
   Divorced/Sep/Widowed  |   .0001626   .0057357     0.03   0.977    -.0110791    .01
          Never Married  |   -.030733    .005813    -5.29   0.000    -.0421263   -.01
---------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.


Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Unemployed), predict(outcome(2))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.fe
               2.metroarea 3.metroarea 1.union 2.marital 3.marital

---------------------------------------------------------------------------------------
                          |            Delta-method
                          |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Inte
--------------------------+------------------------------------------------------------
                   educat |
                     HSD  |   .0021475   .0023962     0.90   0.370     -.002549     .00
            Some College  |  -.0014486   .0026367    -0.55   0.583    -.0066165    .00
           AA/Vocational  |  -.0048071   .0030039    -1.60   0.110    -.0106946    .00
                   BS/BA  |  -.0028255   .0026516    -1.07   0.287    -.0080226    .00
Graduate or Professional  |  -.0081376   .0028311    -2.87   0.004    -.0136864   -.00
                          |
                      exp |   .0004047   .0001592     2.54   0.011     .0000928    .00
                     exp2 |  -.0000155   2.78e-06    -5.57   0.000    -.0000209     -.
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.0176565   .0086621    -2.04   0.042     -.034634   -.00
                   Black  |  -.0099588   .0085704    -1.16   0.245    -.0267565    .00
         Hispanic/Latino  |  -.0128404   .0084135    -1.53   0.127    -.0293307    .00
                   White  |  -.0163498   .0082939    -1.97   0.049    -.0326055   -.00
             Multiracial  |  -.0112902   .0095517    -1.18   0.237    -.0300113    .00
                          |
                   female |
                  Female  |  -.0037756   .0013734    -2.75   0.006    -.0064674   -.00
                          |
                metroarea |
           Nonmetro Area  |  -.0018498   .0017658    -1.05   0.295    -.0053107    .00
```

```
         Not Identified  |   .0012508   .0071177     0.18   0.861    -.0126997    .0152012
                         |
                   union |
                   Union  |  -.0210849   .0007187   -29.34   0.000    -.0224935   -.0196762
                         |
                 marital |
    Divorced/Sep/Widowed  |   .0110753   .0024351     4.55   0.000     .0063027    .0158479
           Never Married  |   .0129138   .0018379     7.03   0.000     .0093115    .0165161
------------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.


Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==NILF), predict(outcome(3))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female
               2.metroarea 3.metroarea 1.union 2.marital 3.marital

------------------------------------------------------------------------------------------
                          |            Delta-method
                          |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
--------------------------+---------------------------------------------------------------
                   educat |
                     HSD  |  -.1330487   .0071954   -18.49   0.000    -.1471514    -.118946
            Some College  |  -.1485456   .0079934   -18.58   0.000    -.1642124   -.1328789
           AA/Vocational  |  -.1863908   .0090294   -20.64   0.000    -.2040881   -.1686934
                   BS/BA  |  -.2379236   .0077818   -30.57   0.000    -.2531757   -.2226716
Graduate or Professional  |  -.2708111   .0083689   -32.36   0.000    -.2872139   -.2544084
                          |
                      exp |  -.0223855   .0004141   -54.06   0.000    -.0231971   -.0215739
                     exp2 |   .0004739   5.54e-06    85.61   0.000     .0004631    .0004848
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.0240837   .0207242    -1.16   0.245    -.0647024    .0165349
                   Black  |  -.0124803   .0200649    -0.62   0.534    -.0518068    .0268463
         Hispanic/Latino  |  -.0673952     .01964    -3.43   0.001    -.1058889   -.0289016
                   White  |  -.0454768   .0191993    -2.37   0.018    -.0831067   -.0078468
             Multiracial  |  -.0235247   .0237295    -0.99   0.322    -.0700337    .0229843
                          |
                   female |
                  Female  |   .0876852   .0038386    22.84   0.000     .0801616    .0952088
                          |
                metroarea |
           Nonmetro Area  |  -.0004278   .0049292    -0.09   0.931    -.0100888    .0092332
```

```
          Not Identified  |  -.0004879   .0192391    -0.03   0.980    -.0381959    .03
                          |
                    union |
                    Union  |   -.420348   .0020016  -210.01   0.000     -.424271   -.41
                          |
                  marital |
     Divorced/Sep/Widowed  |  -.0112379   .0055186    -2.04   0.042    -.0220542   -.00
            Never Married  |   .0178192   .0057407     3.10   0.002     .0065676    .02
--------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

Marginal Effects at the Average

```
margins, dydx(*) atmeans predict(outcome(1))
margins, dydx(*) atmeans predict(outcome(2))
margins, dydx(*) atmeans predict(outcome(3))
```

```
Conditional marginal effects                    Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.fe
               2.metroarea 3.metroarea 1.union 2.marital 3.marital
at           : 1.educat        =    .1261088 (mean)
               2.educat        =     .286954 (mean)
               3.educat        =    .1488497 (mean)
               4.educat        =    .0969254 (mean)
               5.educat        =    .2079617 (mean)
               6.educat        =    .1332003 (mean)
               exp             =    32.89247 (mean)
               exp2            =    1467.595 (mean)
               1.race_eth~y    =    .0098079 (mean)
               2.race_eth~y    =    .0622611 (mean)
               3.race_eth~y    =    .0931032 (mean)
               4.race_eth~y    =    .1487776 (mean)
               5.race_eth~y    =    .6685738 (mean)
               6.race_eth~y    =    .0174764 (mean)
               0.female        =    .4804923 (mean)
               1.female        =    .5195077 (mean)
               1.metroarea     =    .8000673 (mean)
               2.metroarea     =    .1904132 (mean)
               3.metroarea     =    .0095195 (mean)
```

```
0.union       =    .9508642 (mean)
1.union       =    .0491358 (mean)
1.marital     =    .5163586 (mean)
2.marital     =    .1819515 (mean)
3.marital     =    .3016899 (mean)

------------------------------------------------------------------------------------------
                           |            Delta-method
                           |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
---------------------------+--------------------------------------------------------------
                    educat |
                      HSD  |   .1755775   1.437386     0.12   0.903    -2.641647    2.992802
             Some College  |    .196493    1.66501     0.12   0.906    -3.066867    3.459853
            AA/Vocational  |    .240734   2.297249     0.10   0.917     -4.26179    4.743258
                    BS/BA  |   .2905963   3.204011     0.09   0.928     -5.98915    6.570342
Graduate or Professional   |   .3223929   3.782688     0.09   0.932     -7.09154    7.736326
                           |
                       exp |   .0256925   .3928606     0.07   0.948    -.7443001    .7956851
                      exp2 |  -.0005388   .0083342    -0.06   0.948    -.0168735    .0157959
                           |
            race_ethnicity |
  Asian/Pacific Islander   |   .0449909   .7906855     0.06   0.955    -1.504724    1.594706
                    Black  |   .0244574   .4314249     0.06   0.955    -.8211199    .8700347
         Hispanic/Latino   |   .0912758   1.237833     0.07   0.941    -2.334832    2.517383
                    White  |     .06928   .9827052     0.07   0.944    -1.856787    1.995347
              Multiracial  |   .0392659   .5722016     0.07   0.945    -1.082229    1.160761
                           |
                    female |
                   Female  |  -.0981776   1.521079    -0.06   0.949    -3.079437    2.883082
                           |
                 metroarea |
            Nonmetro Area  |   .0018837   .0756332     0.02   0.980    -.1463546     .150122
           Not Identified  |  -.0003745   .0562184    -0.01   0.995    -.1105606    .1098116
                           |
                     union |
                    Union  |    .448807   .0032032   140.11   0.000     .4425288    .4550853
                           |
                   marital |
    Divorced/Sep/Widowed   |   .0044561   .5402485     0.01   0.993    -1.054411    1.063324
           Never Married   |  -.0309979   .6401718    -0.05   0.961    -1.285712    1.223716
------------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.


Conditional marginal effects                    Number of obs     =     41,599
Model VCE    : OIM
```

```
Expression   : Pr(laborforcestatus==Unemployed), predict(outcome(2))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.fe
               2.metroarea 3.metroarea 1.union 2.marital 3.marital
at           : 1.educat       =    .1261088 (mean)
               2.educat       =     .286954 (mean)
               3.educat       =    .1488497 (mean)
               4.educat       =    .0969254 (mean)
               5.educat       =    .2079617 (mean)
               6.educat       =    .1332003 (mean)
               exp            =    32.89247 (mean)
               exp2           =    1467.595 (mean)
               1.race_eth~y   =    .0098079 (mean)
               2.race_eth~y   =    .0622611 (mean)
               3.race_eth~y   =    .0931032 (mean)
               4.race_eth~y   =    .1487776 (mean)
               5.race_eth~y   =    .6685738 (mean)
               6.race_eth~y   =    .0174764 (mean)
               0.female       =    .4804923 (mean)
               1.female       =    .5195077 (mean)
               1.metroarea    =    .8000673 (mean)
               2.metroarea    =    .1904132 (mean)
               3.metroarea    =    .0095195 (mean)
               0.union        =    .9508642 (mean)
               1.union        =    .0491358 (mean)
               1.marital      =    .5163586 (mean)
               2.marital      =    .1819515 (mean)
               3.marital      =    .3016899 (mean)

------------------------------------------------------------------------------------------
                          |            Delta-method
                          |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Inte
--------------------------+---------------------------------------------------------------
                   educat |
                     HSD  |   .0005216   .0755305     0.01   0.994    -.1475154    .148
            Some College  |  -.0012432   .1546272    -0.01   0.994    -.3043069    .301
           AA/Vocational  |  -.0029475   .3436986    -0.01   0.993    -.6765844    .670
                   BS/BA  |  -.0022892   .2747643    -0.01   0.993    -.5408173    .536
Graduate or Professional  |  -.0047363   .5466531    -0.01   0.993    -1.076157    1.06
                          |
                      exp |   .0000394   .0076697     0.01   0.996     -.014993    .015
                     exp2 |  -4.07e-06   .0004725    -0.01   0.993    -.0009301     .00
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.0089668   1.008282    -0.01   0.993    -1.985164     1.9
```

```
                   Black  |  -.0051358   .5753938    -0.01   0.993    -1.132887    1.122615
         Hispanic/Latino  |  -.0069558   .7820821    -0.01   0.993    -1.539809    1.525897
                   White  |  -.0084662   .9520543    -0.01   0.993    -1.874458    1.857526
             Multiracial  |  -.0058748   .6589403    -0.01   0.993    -1.297374    1.285625
                          |
                   female |
                  Female  |  -.0011625   .1324367    -0.01   0.993    -.2607337    .2584087
                          |
               metroarea |
           Nonmetro Area  |  -.0008667   .0980645    -0.01   0.993    -.1930696    .1913361
          Not Identified  |   .0005835   .0659971     0.01   0.993    -.1287684    .1299354
                          |
                   union |
                   Union  |   -.017075   .0009387   -18.19   0.000    -.0189148   -.0152352
                          |
                 marital |
    Divorced/Sep/Widowed  |   .0055891    .631376     0.01   0.993    -1.231885    1.243063
           Never Married  |   .0067698   .7645739     0.01   0.993    -1.491768    1.505307
-------------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.


Conditional marginal effects                    Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==NILF), predict(outcome(3))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat exp exp2 2.race_ethnicity
               3.race_ethnicity 4.race_ethnicity 5.race_ethnicity 6.race_ethnicity 1.female
               2.metroarea 3.metroarea 1.union 2.marital 3.marital
at           : 1.educat       =    .1261088 (mean)
               2.educat       =     .286954 (mean)
               3.educat       =    .1488497 (mean)
               4.educat       =    .0969254 (mean)
               5.educat       =    .2079617 (mean)
               6.educat       =    .1332003 (mean)
               exp            =    32.89247 (mean)
               exp2           =    1467.595 (mean)
               1.race_eth~y   =    .0098079 (mean)
               2.race_eth~y   =    .0622611 (mean)
               3.race_eth~y   =    .0931032 (mean)
               4.race_eth~y   =    .1487776 (mean)
               5.race_eth~y   =    .6685738 (mean)
               6.race_eth~y   =    .0174764 (mean)
               0.female       =    .4804923 (mean)
               1.female       =    .5195077 (mean)
               1.metroarea    =    .8000673 (mean)
```

```
               2.metroarea     =    .1904132 (mean)
               3.metroarea     =    .0095195 (mean)
               0.union         =    .9508642 (mean)
               1.union         =    .0491358 (mean)
               1.marital       =    .5163586 (mean)
               2.marital       =    .1819515 (mean)
               3.marital       =    .3016899 (mean)

------------------------------------------------------------------------------------
                          |            Delta-method
                          |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Inte
--------------------------+---------------------------------------------------------
                   educat |
                      HSD |  -.1760991   1.475242   -0.12   0.905     -3.06752    2.7
             Some College |  -.1952498   1.746298   -0.11   0.911    -3.617932    3.2
            AA/Vocational |  -.2377865   2.409215   -0.10   0.921    -4.959762    4.4
                    BS/BA |  -.2883071   3.317729   -0.09   0.931    -6.790936    6.2
Graduate or Professional  |  -.3176566   3.916079   -0.08   0.935    -7.993031    7.3
                          |
                      exp |  -.0257319   .3987801   -0.06   0.949    -.8073266    .75
                     exp2 |   .0005428   .0084077    0.06   0.949    -.0159359    .01
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.0360241   .5758254   -0.06   0.950    -1.164621    1.0
                    Black |  -.0193216   .3135138   -0.06   0.951    -.6337973    .59
          Hispanic/Latino |    -.08432   1.269837   -0.07   0.947    -2.573155    2.4
                    White |  -.0608138   .9093883   -0.07   0.947    -1.843182    1.7
              Multiracial |  -.0333911     .49753   -0.07   0.946    -1.008532    .94
                          |
                   female |
                   Female |   .0993401   1.526768    0.07   0.948     -2.89307    3.0
                          |
                metroarea |
            Nonmetro Area |   -.001017   .0285739   -0.04   0.972    -.0570207    .05
           Not Identified |  -.0002091   .0267367   -0.01   0.994     -.052612    .05
                          |
                    union |
                    Union |   -.431732   .0031813 -135.71   0.000    -.4379673   -.42
                          |
                  marital |
     Divorced/Sep/Widowed |  -.0100452   .2043758   -0.05   0.961    -.4106144     .3
            Never Married |   .0242282    .419673    0.06   0.954    -.7983157     .8
------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

Change the Base Alternative

We can also change the base or reference alternative with the base() option

```
mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.marital, base
```

```
Iteration 0:   log likelihood = -31774.039
Iteration 1:   log likelihood = -22976.886
Iteration 2:   log likelihood = -22621.724
Iteration 3:   log likelihood = -22574.602
Iteration 4:   log likelihood = -22567.151
Iteration 5:   log likelihood = -22565.706
Iteration 6:   log likelihood = -22565.376
Iteration 7:   log likelihood = -22565.303
Iteration 8:   log likelihood = -22565.287
Iteration 9:   log likelihood = -22565.285
Iteration 10:  log likelihood = -22565.284
Iteration 11:  log likelihood = -22565.284
Iteration 12:  log likelihood = -22565.284

Multinomial logistic regression                 Number of obs     =     41,599
                                                LR chi2(36)       =   18417.51
                                                Prob > chi2       =     0.0000
Log likelihood = -22565.284                     Pseudo R2         =     0.2898

------------------------------------------------------------------------------------------
         laborforcestatus |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
--------------------------+---------------------------------------------------------------
Employed                  |
                   educat |
                     HSD  |   .7894788   .0425405    18.56   0.000     .7061009    .8728567
            Some College  |   .8934956   .0479536    18.63   0.000     .7995083    .9874829
           AA/Vocational  |   1.142766    .055907    20.44   0.000      1.03319    1.252341
                   BS/BA  |   1.492912   .0488292    30.57   0.000     1.397209    1.588616
Graduate or Professional  |   1.746553    .056187    31.08   0.000     1.636428    1.856677
                          |
                      exp |   .1490952   .0030868    48.30   0.000     .1430452    .1551451
                     exp2 |  -.0031411   .0000469   -66.92   0.000    -.0032331   -.0030491
                          |
           race_ethnicity |
  Asian/Pacific Islander  |    .199411   .1336958     1.49   0.136    -.062628      .46145
                   Black  |   .1055321   .1293351     0.82   0.415     -.14796    .3590242
         Hispanic/Latino  |   .4793179   .1270317     3.77   0.000     .2303404    .7282954
                   White  |   .3381891    .123776     2.73   0.006     .0955926    .5807856
             Multiracial  |   .1810197   .1537766     1.18   0.239    -.1203769    .4824163
```

```
                          |
                   female |
                  Female  |  -.5781724   .0258932   -22.33   0.000    -.6289222   -.527
                          |
                metroarea |
            Nonmetro Area |    .007027   .0328055     0.21   0.830    -.0572707    .07
           Not Identified |   .0004518   .1282484     0.00   0.997    -.2509105    .25
                          |
                    union |
                   Union  |   20.25617   567.0506     0.04   0.972    -1091.143    113
                          |
                  marital |
     Divorced/Sep/Widowed |   .0530922   .0373158     1.42   0.155    -.0200455     .
            Never Married |  -.1465344   .0379554    -3.86   0.000    -.2209255   -.072
                          |
                    _cons |  -1.052655   .1359142    -7.74   0.000    -1.319042   -.786
--------------------------+----------------------------------------------------------------
Unemployed                |
                   educat |
                     HSD  |     .56128   .1124074     4.99   0.000     .3409657    .78
             Some College |   .4618071   .1313983     3.51   0.000     .2042712     .7
            AA/Vocational |   .4492192   .1680784     2.67   0.008     .1197916    .778
                    BS/BA |   .8221872   .1354557     6.07   0.000     .5566988    1.08
Graduate or Professional  |   .6830694   .1777718     3.84   0.000      .334643    1.03
                          |
                      exp |   .1200086   .0089721    13.38   0.000     .1024235    .137
                     exp2 |  -.0028889   .0001584   -18.24   0.000    -.0031994   -.002
                          |
           race_ethnicity |
  Asian/Pacific Islander  |  -.6146734   .3071598    -2.00   0.045    -1.216696   -.012
                   Black  |  -.2893004   .2809389    -1.03   0.303    -.8399305    .26
         Hispanic/Latino  |  -.1705519   .2744134    -0.62   0.534    -.7083923    .367
                   White  |  -.4471628   .2665893    -1.68   0.093    -.9696683    .075
             Multiracial  |  -.2985012    .342879    -0.87   0.384    -.9705317    .373
                          |
                   female |
                  Female  |  -.5787555   .0737225    -7.85   0.000    -.7232489    -.43
                          |
                metroarea |
            Nonmetro Area |  -.0940971   .0993931    -0.95   0.344    -.2889041    .100
           Not Identified |   .0622817   .3526626     0.18   0.860    -.6289243    .753
                          |
                    union |
                   Union  |   .5900779   2407.697     0.00   1.000    -4718.409    471
                          |
                  marital |
```

```
    Divorced/Sep/Widowed  |   .6663185   .1197798     5.56   0.000     .4315545    .9010826
           Never Married  |   .6061848   .1022777     5.93   0.000     .4057242    .8066454
                          |
                   _cons  |  -3.064479    .303667   -10.09   0.000    -3.659655   -2.469302
--------------------------+----------------------------------------------------------------
NILF                      |  (base outcome)
-------------------------------------------------------------------------------------------
Note: 2044 observations completely determined.  Standard errors questionable.
```

Marginal Effects and Marginsplot

Next, we will estimate and graph the average marginal effects for education and experience

```
quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.metroarea i.union i.marit
```

Education

For Employed

```
margins, dydx(educat) predict(outcome(1))
quietly marginsplot, allsimplelabels horizontal recast(scatter) name(Employed) yscale(reverse) yt
```

```
Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : 2.educat 3.educat 4.educat 5.educat 6.educat

-------------------------------------------------------------------------------------------
                          |            Delta-method
                          |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
--------------------------+----------------------------------------------------------------
                   educat |
                     HSD  |   .1309012   .0072675    18.01   0.000     .1166572    .1451452
            Some College  |   .1499942   .0080928    18.53   0.000     .1341326    .1658558
           AA/Vocational  |   .1911979   .0091537    20.89   0.000     .1732569    .2091389
                   BS/BA  |   .2407492   .0078942    30.50   0.000     .2252768    .2562215
Graduate or Professional  |   .2789487    .008527    32.71   0.000     .2622362    .2956613
-------------------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

Individuals with high school degree have 13.1 percentage point more likely to be employed compared to high school dropouts. Individuals with some college are 15 percentage points more likely to be employed compared to high school dropouts. Individuals with Associates or Vocational degrees are 19.1 percentage points more likely to be employed compared to high school dropouts. Individuals with a Bachelor's degree are 24.1 percentage points more likely to be employed, while individuals with a graduate degree are 27.9 percentage points more likely to be employed compared to high school dropouts.

For Unemployed

```
margins, dydx(educat) predict(outcome(2))
quietly marginsplot, allsimplelabels horizontal recast(scatter) name(Unemployed) yscal
```

For Not in the Labor Force

```
margins, dydx(educat) predict(outcome(3))
quilety marginsplot, allsimplelabels horizontal recast(scatter) name(NILF) yscale(reve
```

Combine graphs

```
graph combine Employed Unemployed NILF, ycommon title("AME of Education") ///
note("Source: Current Population Survey; 2 is HSD, 3 is Some College, 4 is AA, 5 is BA
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmeducation.png", replace
```

```
graph drop Employed Unemployed NILF
```

Potential Experience

For Employed

```
margins, dydx(exp) at(exp=(0(2)60)) predict(outcome(1))
quietly marginsplot, allsimplelabels name(Employed) ytitle("Effect on Pr(Employed)") x
```

```
Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==Employed), predict(outcome(1))
dy/dx w.r.t. : exp

1._at        : exp             =           0

2._at        : exp             =           2
```

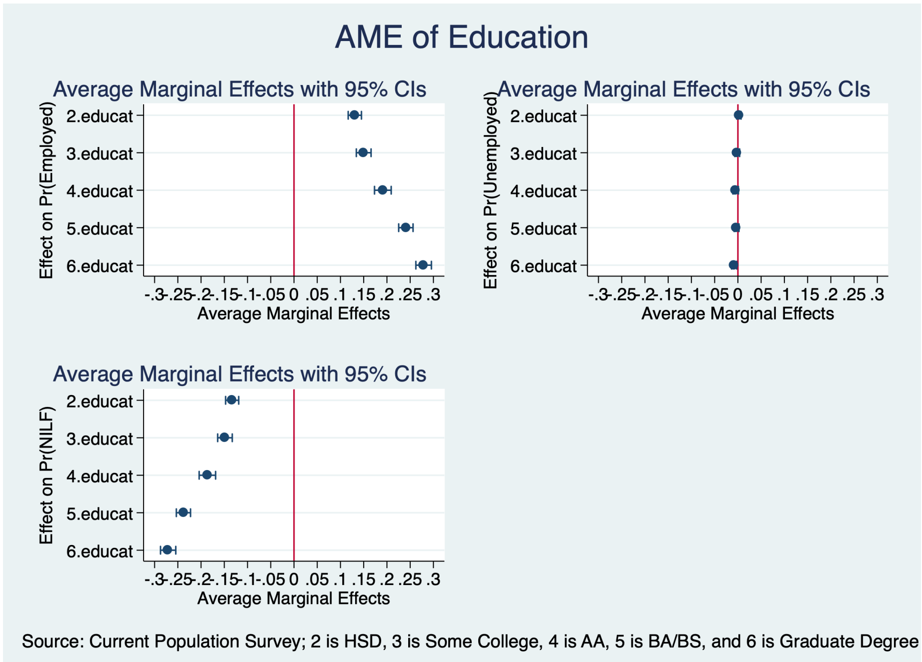

Figure 2.1: Marginal Effects of Education

```
3._at          : exp              =          4

4._at          : exp              =          6

5._at          : exp              =          8

6._at          : exp              =         10

7._at          : exp              =         12

8._at          : exp              =         14

9._at          : exp              =         16

10._at         : exp              =         18

11._at         : exp              =         20

12._at         : exp              =         22

13._at         : exp              =         24
```

```
14._at       : exp             =          26

15._at       : exp             =          28

16._at       : exp             =          30

17._at       : exp             =          32

18._at       : exp             =          34

19._at       : exp             =          36

20._at       : exp             =          38

21._at       : exp             =          40

22._at       : exp             =          42

23._at       : exp             =          44

24._at       : exp             =          46

25._at       : exp             =          48

26._at       : exp             =          50

27._at       : exp             =          52

28._at       : exp             =          54

29._at       : exp             =          56

30._at       : exp             =          58

31._at       : exp             =          60

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
exp          |
         _at |
          1  |   .0119199   .0001127   105.73   0.000     .0116989    .0121408
          2  |   .0130781   .0001155   113.28   0.000     .0128519    .0133044
          3  |    .013976   .0001461    95.64   0.000     .0136896    .0142625
```

```
        4  |   .0145643   .0001796    81.10   0.000     .0142123    .0149163
        5  |   .0148298   .0002035    72.86   0.000     .0144308    .0152287
        6  |   .0147942    .000215    68.83   0.000     .0143729    .0152155
        7  |   .0145067   .0002152    67.40   0.000     .0140849    .0149285
        8  |   .0140316   .0002075    67.61   0.000     .0136249    .0144384
        9  |   .0134367   .0001953    68.81   0.000      .013054    .0138195
       10  |   .0127838   .0001811    70.60   0.000     .0124289    .0131387
       11  |   .0121231   .0001667    72.73   0.000     .0117965    .0124498
       12  |   .0114913    .000153    75.12   0.000     .0111915    .0117912
       13  |   .0109121   .0001403    77.76   0.000     .0106371    .0111872
       14  |   .0103981   .0001289    80.69   0.000     .0101455    .0106507
       15  |   .0099532   .0001186    83.95   0.000     .0097208    .0101855
       16  |   .0095751   .0001094    87.52   0.000     .0093606    .0097895
       17  |   .0092574   .0001014    91.30   0.000     .0090586    .0094561
       18  |   .0089913   .0000945    95.12   0.000      .008806    .0091766
       19  |   .0087672   .0000887    98.80   0.000     .0085933    .0089411
       20  |   .0085751   .0000839   102.22   0.000     .0084107    .0087395
       21  |   .0084059   .0000798   105.34   0.000     .0082495    .0085623
       22  |   .0082513   .0000762   108.24   0.000     .0081019    .0084007
       23  |   .0081038   .0000729   111.13   0.000     .0079609    .0082468
       24  |   .0079576   .0000697   114.23   0.000      .007821    .0080941
       25  |   .0078075   .0000663   117.83   0.000     .0076776    .0079373
       26  |   .0076498   .0000626   122.19   0.000     .0075271    .0077725
       27  |   .0074817   .0000586   127.62   0.000     .0073668    .0075967
       28  |   .0073016   .0000543   134.46   0.000     .0071952    .0074081
       29  |   .0071087   .0000497   143.16   0.000     .0070113     .007206
       30  |   .0069031   .0000447   154.27   0.000     .0068154    .0069908
       31  |   .0066862   .0000397   168.35   0.000     .0066084    .0067641
------------------------------------------------------------------------------
```

For Not in the Labor Force

```
margins, dydx(exp) at(exp=(0(2)60)) predict(outcome(3))
quietly marginsplot, allsimplelabels name(NILF) ytitle("Effect on Pr(NILF)") xtitle("Average Marg
```

```
Average marginal effects                        Number of obs     =     41,599
Model VCE    : OIM

Expression   : Pr(laborforcestatus==NILF), predict(outcome(3))
dy/dx w.r.t. : exp

1._at        : exp             =           0

2._at        : exp             =           2
```

```
3._at          : exp             =          4

4._at          : exp             =          6

5._at          : exp             =          8

6._at          : exp             =         10

7._at          : exp             =         12

8._at          : exp             =         14

9._at          : exp             =         16

10._at         : exp             =         18

11._at         : exp             =         20

12._at         : exp             =         22

13._at         : exp             =         24

14._at         : exp             =         26

15._at         : exp             =         28

16._at         : exp             =         30

17._at         : exp             =         32

18._at         : exp             =         34

19._at         : exp             =         36

20._at         : exp             =         38

21._at         : exp             =         40

22._at         : exp             =         42

23._at         : exp             =         44

24._at         : exp             =         46

25._at         : exp             =         48
```

```
26._at       : exp             =          50

27._at       : exp             =          52

28._at       : exp             =          54

29._at       : exp             =          56

30._at       : exp             =          58

31._at       : exp             =          60

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
exp          |
         _at |
          1  |  -.0125848   .0001155  -108.97   0.000    -.0128111   -.0123584
          2  |  -.0137412   .0001266  -108.55   0.000    -.0139893    -.013493
          3  |  -.0146118    .000159   -91.88   0.000    -.0149235   -.0143001
          4  |  -.0151489   .0001885   -80.35   0.000    -.0155185   -.0147794
          5  |  -.0153436   .0002048   -74.91   0.000     -.015745   -.0149421
          6  |  -.0152237   .0002061   -73.87   0.000    -.0156276   -.0148198
          7  |  -.0148454   .0001949   -76.17   0.000    -.0152274   -.0144634
          8  |  -.0142796   .0001756   -81.30   0.000    -.0146238   -.0139353
          9  |  -.0135992   .0001528   -89.00   0.000    -.0138986   -.0132997
         10  |  -.0128695   .0001299   -99.06   0.000    -.0131241   -.0126148
         11  |  -.0121426   .0001093  -111.07   0.000    -.0123569   -.0119284
         12  |  -.0114558   .0000922  -124.29   0.000    -.0116365   -.0112752
         13  |  -.0108322   .0000787  -137.62   0.000    -.0109865   -.0106779
         14  |  -.0102833   .0000686  -149.81   0.000    -.0104178   -.0101487
         15  |  -.0098117   .0000614  -159.70   0.000    -.0099321   -.0096913
         16  |  -.0094137   .0000566  -166.42   0.000    -.0095246   -.0093028
         17  |  -.0090816   .0000536  -169.49   0.000    -.0091866   -.0089766
         18  |  -.0088055   .0000521  -169.01   0.000    -.0089076   -.0087033
         19  |  -.0085745   .0000517  -165.70   0.000    -.0086759   -.0084731
         20  |   -.008378   .0000521  -160.73   0.000    -.0084802   -.0082759
         21  |  -.0082063   .0000528  -155.31   0.000    -.0083098   -.0081027
         22  |  -.0080503   .0000535  -150.38   0.000    -.0081552   -.0079453
         23  |  -.0079024   .0000539  -146.55   0.000    -.0080081   -.0077967
         24  |  -.0077563   .0000538  -144.13   0.000    -.0078617   -.0076508
         25  |  -.0076067   .0000531  -143.27   0.000    -.0077108   -.0075027
         26  |  -.0074499   .0000517  -144.07   0.000    -.0075512   -.0073485
         27  |  -.0072829   .0000497  -146.60   0.000    -.0073802   -.0071855
         28  |   -.007104    .000047  -151.07   0.000    -.0071962   -.0070118
```

```
         29  |  -.0069124   .0000438  -157.78   0.000    -.0069983   -.0068266
         30  |  -.0067085   .0000401  -167.14   0.000    -.0067872   -.0066298
         31  |  -.0064935   .0000362  -179.60   0.000    -.0065643   -.0064226
------------------------------------------------------------------------------
```

Combine Graphs

```
graph combine Employed NILF, ycommon title("AME of Potential Experience")
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmexp.png", replace
```

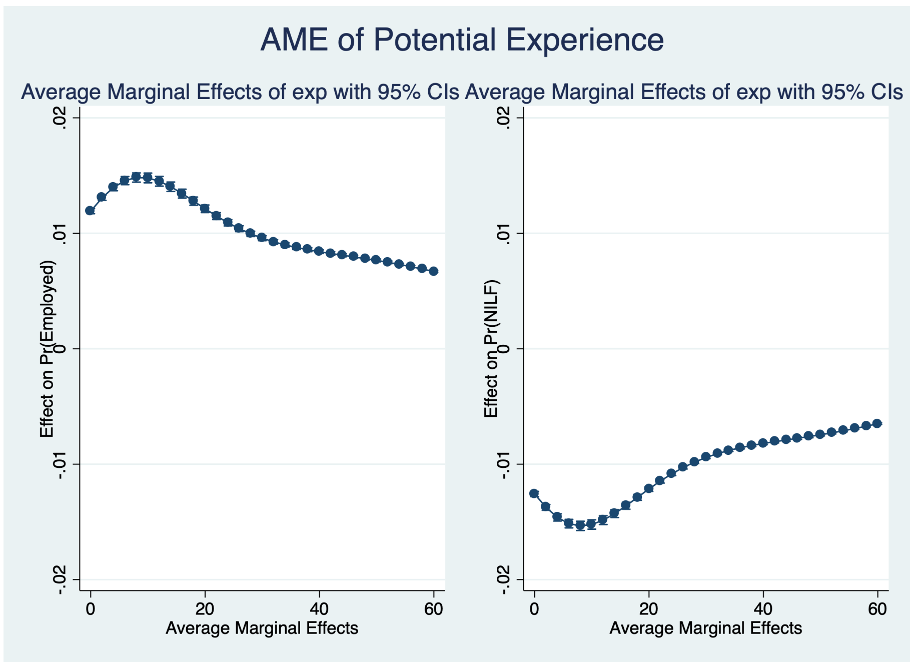

Figure 2.2: Marginal Effects of Potential Experience

```
graph drop Employed NILF
```

We can use coefplot with margins with eststo

```
quilety {est clear
eststo mnlm: quietly mlogit laborforce i.educat exp exp2 i.race_ethnicity i.female i.me
}
```

Estimate the average marginal effects. Please note the post option when storing margin results

```
quietly{
eststo Employed: quietly margins, dydx(educat) predict(outcome(1)) post
estimates restore mnlm
eststo Unemployed: quietly margins, dydx(educat) predict(outcome(2)) post
estimates restore mnlm
eststo NILF: quietly margins, dydx(educat) predict(outcome(3)) post
}
```

Use coefplot

```
coefplot Employed Unemployed NILF, ///
recast(bar) barw(0.15) vertical ///
ciopts(recast(rcap) color(gs8)) citop ///
xlab(1 "High School" 2 "Some College" 3 "Associates" 4 "Bachelor" 5 "Graduate") ///
ytitle("Average Marginal Effect") ///
xtitle("High Level of Education Attained") ///
title("MNLM: Probability of Labor Force Status", size(*0.7))    ///
subtitle("By Education Relative to High School Dropout")    ///
caption("Source: Current Population Survey", size(*0.75)) ///
name(coefplot3)
graph export "/Users/Sam/Desktop/Econ 645/Stata/week8_mnlmcoefplot.png", replace
```

```
graph drop coefplot3
```

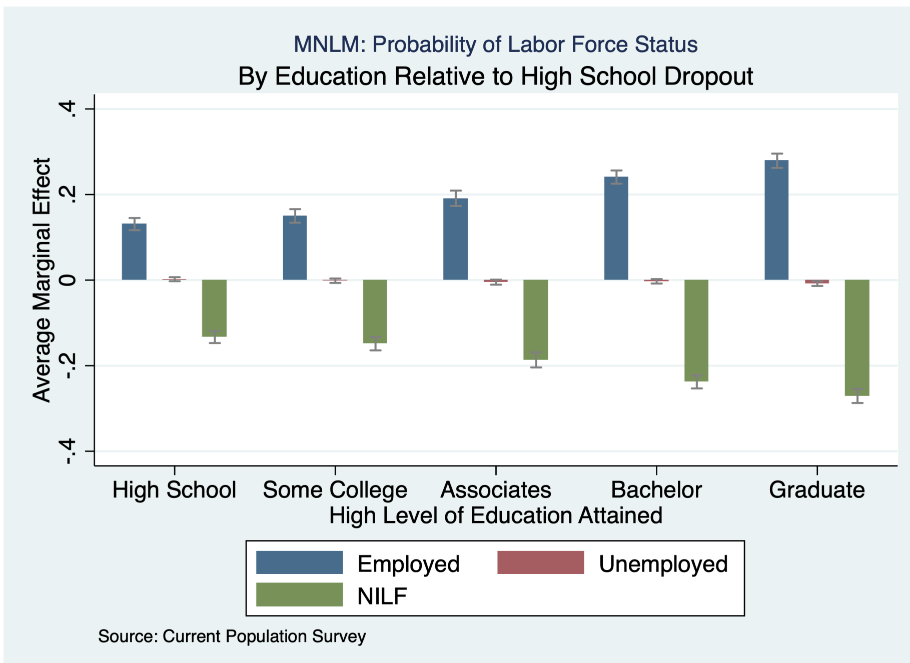

Figure 2.3: Using coefplot instead of marginsplot

Chapter 3

Ordinal Logit

Adapted from Long and Freese (2006)

3.1 Compare Ordinal Logit and Binary Logit

First, let's inspect the outcome of interest, which is labor force status.

```
use "/Users/Sam/Desktop/Econ 645/Data/binlfp2.dta", clear
tab lfp
```

```
(Data from 1976 PSID-T Mroz)


 Paid Labor |
     Force: |
 1=yes 0=no |      Freq.     Percent        Cum.
------------+-----------------------------------
    NotInLF |        325       43.16       43.16
       inLF |        428       56.84      100.00
------------+-----------------------------------
      Total |        753      100.00
```

Now, let's compare our ordinal logit and binary logit coefficients when the outcome is binary.

```
quietly logit lfp k5 k618 age wc hc lwg inc, nolog
estimates store logit
quietly ologit lfp k5 k618 age wc hc lwg inc, nolog
```

```
estimates store ologit
estimates table logit ologit, b(%9.3f) t varlabel varwidth(30) equations(1:1)
```

```
                      Variable |   logit      ologit
-------------------------------+------------------------
#1                             |
                    # kids < 6 |    -1.463      -1.463
                               |     -7.43       -7.43
                   # kids 6-18 |    -0.065      -0.065
                               |     -0.95       -0.95
             Wife's age in years |    -0.063      -0.063
                               |     -4.92       -4.92
       Wife College: 1=yes 0=no |     0.807       0.807
                               |      3.51        3.51
    Husband College: 1=yes 0=no |     0.112       0.112
                               |      0.54        0.54
  Log of wife's estimated wages |     0.605       0.605
                               |      4.01        4.01
Family income excluding wife's |    -0.034      -0.034
                               |     -4.20       -4.20
                      Constant |     3.182
                               |      4.94
-------------------------------+------------------------
cut1                           |
                      Constant |                -3.182
                               |                 -4.94
--------------------------------------------------------
                                             legend: b/t
```

The coefficients are the same between the binary logit and the ordinal logit. The difference here is that the ordinal logit has a cut point and assumes an interept of 0. The cutpoint and the intercept for binary logit are the same in absolute terms.

3.2 Example of Ordinal Dependent Variable

```
use "/Users/Sam/Desktop/Econ 645/Data/ordwarm2.dta", clear
describe warm yr89 male white age ed prst
```

```
(77 & 89 General Social Survey)
```

```
              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------------------
warm            byte    %10.0g     SD2SA      Mom can have warm relations with child
yr89            byte    %10.0g     yrlbl      Survey year: 1=1989 0=1977
male            byte    %10.0g     sexlbl     Gender: 1=male 0=female
white           byte    %10.0g     racelbl    Race: 1=white 0=not white
age             byte    %10.0g                Age in years
ed              byte    %10.0g                Years of education
prst            byte    %10.0g                Occupational prestige
```

```
summarize warm yr89 male white age ed prst
```

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
        warm |      2,293    2.607501    .9282156          1          4
        yr89 |      2,293    .3986044    .4897178          0          1
        male |      2,293    .4648932    .4988748          0          1
       white |      2,293    .8765809    .3289894          0          1
         age |      2,293    44.93546    16.77903         18         89
-------------+---------------------------------------------------------
          ed |      2,293    12.21805    3.160827          0         20
        prst |      2,293    39.58526    14.49226         12         82
```

Our outcome is the the survey results for people's opinion for a working mother

```
tab warm
```

```
    Mom can |
  have warm |
  relations |
 with child |      Freq.     Percent        Cum.
------------+-----------------------------------
         SD |        297       12.95       12.95
          D |        723       31.53       44.48
          A |        856       37.33       81.81
         SA |        417       18.19      100.00
------------+-----------------------------------
      Total |      2,293      100.00
```

We will run our ordinal logit by regressing opinion's on working mother's warm onto gender, race, age, survey year, education in years, and occupational prestige.

$$warm_{i,t} = \beta_1 male_i + \beta_2 race_i + \beta_3 age_{i,t} + \beta_4 edu_{i,t} + \beta_5 prestige_{i,t} + \beta_6 y89_t + \varepsilon_{i,t}$$

```
ologit warm yr89 male white age ed prst, nolog
```

```
Ordered logistic regression                     Number of obs     =      2,293
                                                LR chi2(6)        =     301.72
                                                Prob > chi2       =     0.0000
Log likelihood = -2844.9123                     Pseudo R2         =     0.0504

------------------------------------------------------------------------------
        warm |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   .5239025   .0798989     6.56   0.000     .3673036    .6805014
        male |  -.7332997   .0784827    -9.34   0.000     -.887123   -.5794765
       white |  -.3911595   .1183808    -3.30   0.001    -.6231816   -.1591373
         age |  -.0216655   .0024683    -8.78   0.000    -.0265032   -.0168278
          ed |   .0671728    .015975     4.20   0.000     .0358624    .0984831
        prst |   .0060727   .0032929     1.84   0.065    -.0003813    .0125267
-------------+----------------------------------------------------------------
       /cut1 |  -2.465362   .2389128                     -2.933622   -1.997102
       /cut2 |   -.630904   .2333156                     -1.088194   -.1736138
       /cut3 |   1.261854    .234018                      .8031871    1.720521
------------------------------------------------------------------------------
```

We have three cutpoints

$$\tau_1 = -2.47, \tau_2 = -0.63, \tau_3 = 1.26$$

3.3 Test Proportional Odds Assumption

We have our two tests for testing the proportional odds assumption 1. Approximate Likelihood Ratio Test 2. Wald Test

3.3.1 Approximate Likelihood Ratio Test

We use the *omodel* command to run an Approximate Likelihood Ratio Test

```
omodel logit warm yr89 male white age ed prst
```

```
Iteration 0:   log likelihood = -2995.7704
Iteration 1:   log likelihood = -2846.4532
Iteration 2:   log likelihood = -2844.9142
Iteration 3:   log likelihood = -2844.9123

Ordered logit estimates                           Number of obs   =       2293
                                                  LR chi2(6)      =     301.72
                                                  Prob > chi2     =     0.0000
Log likelihood = -2844.9123                       Pseudo R2       =     0.0504

------------------------------------------------------------------------------
        warm |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   .5239025   .0798988     6.56   0.000     .3673037    .6805013
        male |  -.7332997   .0784827    -9.34   0.000    -.8871229   -.5794766
       white |  -.3911595   .1183808    -3.30   0.001    -.6231815   -.1591374
         age |  -.0216655   .0024683    -8.78   0.000    -.0265032   -.0168278
          ed |   .0671728    .015975     4.20   0.000     .0358624    .0984831
        prst |   .0060727   .0032929     1.84   0.065    -.0003813    .0125267
-------------+----------------------------------------------------------------
       _cut1 |  -2.465362   .2389126          (Ancillary parameters)
       _cut2 |   -.630904   .2333155
       _cut3 |   1.261854   .2340179
------------------------------------------------------------------------------

Approximate likelihood-ratio test of proportionality of odds
across response categories:
         chi2(12) =     48.91
       Prob > chi2 =    0.0000
```

With the oparallel package

```
help oparallel
quietly omodel logit warm yr89 male white age ed prst
oparallel, lr
```

```
Test of the parallel regression assumption

                 |   Chi2     df  P>Chi2
-----------------+----------------------
likelihood ratio |   49.2     12   0.000
```

We reject the null hypothesis that the coefficients are the same across $1, ..., J-1$ equations at the 0.1% level.

3.3.2 Wald Test

We use the *brant* command for the Wald Test

```
quietly ologit warm yr89 male white age ed prst, nolog
oparallel, lr
```

```
quietly use "/Users/Sam/Desktop/Econ 645/Data/ordwarm2.dta", clear
quietly ologit warm yr89 male white age ed prst, nolog
oparallel, lr
```

We reject the null hypothesis that the coefficients are the same across $1, ..., J-1$ equations at the 0.1% level.

3.4 Interpretation

3.4.1 Odds Ratios

```
ologit warm yr89 male white age ed prst, or
oparallel, lr
listcoef, help
listcoef, help percent
```

```
Iteration 0:   log likelihood = -2995.7704
Iteration 1:   log likelihood = -2846.4532
Iteration 2:   log likelihood = -2844.9142
Iteration 3:   log likelihood = -2844.9123
Iteration 4:   log likelihood = -2844.9123

Ordered logistic regression                     Number of obs     =      2,293
                                                LR chi2(6)        =     301.72
                                                Prob > chi2       =     0.0000
Log likelihood = -2844.9123                     Pseudo R2         =     0.0504

------------------------------------------------------------------------------
        warm | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   1.688605   .1349176     6.56   0.000     1.443836    1.974868
        male |   .4803214   .0376969    -9.34   0.000     .4118389    .5601916
       white |   .6762723   .0800577    -3.30   0.001     .5362356    .8528792
         age |   .9785675   .0024154    -8.78   0.000     .9738449     .983313
          ed |    1.06948   .0170849     4.20   0.000     1.036513    1.103496
```

```
        prst |   1.006091    .003313     1.84   0.065     .9996188    1.012605
-------------+----------------------------------------------------------------
       /cut1 |  -2.465362   .2389128                     -2.933622   -1.997102
       /cut2 |   -.630904   .2333156                     -1.088194   -.1736138
       /cut3 |   1.261854    .234018                      .8031871    1.720521
------------------------------------------------------------------------------


ologit (N=2293): Factor Change in Odds

  Odds of: >m vs <=m

------------------------------------------------------------------
    warm |      b         z     P>|z|    e^b    e^bStdX      SDofX
---------+--------------------------------------------------------
    yr89 |   0.52390    6.557   0.000   1.6886   1.2925     0.4897
    male |  -0.73330   -9.343   0.000   0.4803   0.6936     0.4989
   white |  -0.39116   -3.304   0.001   0.6763   0.8792     0.3290
     age |  -0.02167   -8.778   0.000   0.9786   0.6952    16.7790
      ed |   0.06717    4.205   0.000   1.0695   1.2365     3.1608
    prst |   0.00607    1.844   0.065   1.0061   1.0920    14.4923
------------------------------------------------------------------
       b = raw coefficient
       z = z-score for test of b=0
   P>|z| = p-value for z-test
     e^b = exp(b) = factor change in odds for unit increase in X
 e^bStdX = exp(b*SD of X) = change in odds for SD increase in X
   SDofX = standard deviation of X


ologit (N=2293): Percentage Change in Odds

  Odds of: >m vs <=m

------------------------------------------------------------------
    warm |      b         z     P>|z|      %      %StdX      SDofX
---------+--------------------------------------------------------
    yr89 |   0.52390    6.557   0.000     68.9     29.2     0.4897
    male |  -0.73330   -9.343   0.000    -52.0    -30.6     0.4989
   white |  -0.39116   -3.304   0.001    -32.4    -12.1     0.3290
     age |  -0.02167   -8.778   0.000     -2.1    -30.5    16.7790
      ed |   0.06717    4.205   0.000      6.9     23.7     3.1608
    prst |   0.00607    1.844   0.065      0.6      9.2    14.4923
------------------------------------------------------------------
       b = raw coefficient
       z = z-score for test of b=0
```

```
P>|z| = p-value for z-test
    % = percent change in odds for unit increase in X
%StdX = percent change in odds for SD increase in X
SDofX = standard deviation of X
```

For a 1-year increase in education , the odds of a **positive attitude** towards working mothers increase by a factor of 1.067 or 6.7%, ceteris paribus.

For a 1-year increase in age, the odds of a **negative** attitude toward working mothers increase by a factor of $e^{-(-0.02167)} = 1.0219$, ceteris paribus.

For a male, the odds of a **negative attitude** toward working mothers increase by a factor of $e^{-(-0.73330)} = 2.08$, ceteris paribus

3.4.2 Average Marginal Effects

```
ologit warm yr89 male white age ed prst, or

margins, dydx(*) predict(outcome(1))
margins, dydx(*) predict(outcome(2))
margins, dydx(*) predict(outcome(3))
margins, dydx(*) predict(outcome(4))
```

```
Iteration 0:   log likelihood = -2995.7704
Iteration 1:   log likelihood = -2846.4532
Iteration 2:   log likelihood = -2844.9142
Iteration 3:   log likelihood = -2844.9123
Iteration 4:   log likelihood = -2844.9123

Ordered logistic regression                     Number of obs     =      2,293
                                                LR chi2(6)        =     301.72
                                                Prob > chi2       =     0.0000
Log likelihood = -2844.9123                     Pseudo R2         =     0.0504

------------------------------------------------------------------------------
        warm | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   1.688605   .1349176     6.56   0.000     1.443836    1.974868
        male |   .4803214   .0376969    -9.34   0.000     .4118389    .5601916
       white |   .6762723   .0800577    -3.30   0.001     .5362356    .8528792
         age |   .9785675   .0024154    -8.78   0.000     .9738449     .983313
          ed |    1.06948   .0170849     4.20   0.000     1.036513    1.103496
        prst |   1.006091    .003313     1.84   0.065     .9996188    1.012605
-------------+----------------------------------------------------------------
```

```
       /cut1 |  -2.465362   .2389128                     -2.933622   -1.997102
       /cut2 |   -.630904   .2333156                     -1.088194   -.1736138
       /cut3 |   1.261854    .234018                      .8031871    1.720521
------------------------------------------------------------------------------


Average marginal effects                        Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==1), predict(outcome(1))
dy/dx w.r.t. : yr89 male white age ed prst

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |  -.0557094   .0087921    -6.34   0.000    -.0729416   -.0384771
        male |   .0779757   .0088072     8.85   0.000     .0607139    .0952375
       white |   .0415941   .0127155     3.27   0.001     .0166721    .0665161
         age |   .0023038   .0002754     8.37   0.000     .0017641    .0028435
          ed |  -.0071428   .0017147    -4.17   0.000    -.0105036   -.0037821
        prst |  -.0006457   .0003514    -1.84   0.066    -.0013345     .000043
------------------------------------------------------------------------------


Average marginal effects                        Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==2), predict(outcome(2))
dy/dx w.r.t. : yr89 male white age ed prst

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |  -.0608201   .0091774    -6.63   0.000    -.0788075   -.0428328
        male |   .0851292   .0089196     9.54   0.000     .0676472    .1026112
       white |   .0454099   .0136969     3.32   0.001     .0185644    .0722554
         age |   .0025152   .0002817     8.93   0.000      .001963    .0030673
          ed |  -.0077981     .00185    -4.22   0.000    -.0114241   -.0041722
        prst |   -.000705   .0003818    -1.85   0.065    -.0014533    .0000433
------------------------------------------------------------------------------


Average marginal effects                        Number of obs     =      2,293
Model VCE    : OIM
```

```
Expression   : Pr(warm==3), predict(outcome(3))
dy/dx w.r.t. : yr89 male white age ed prst

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |    .043531   .0069122     6.30   0.000     .0299833    .0570787
        male |  -.0609298   .0067839    -8.98   0.000     -.074226   -.0476336
       white |  -.0325014   .0099733    -3.26   0.001    -.0520487   -.0129541
         age |  -.0018002   .0002138    -8.42   0.000    -.0022193   -.0013811
          ed |   .0055814   .0013342     4.18   0.000     .0029665    .0081963
        prst |   .0005046   .0002752     1.83   0.067    -.0000349    .0010441
------------------------------------------------------------------------------


Average marginal effects                        Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==4), predict(outcome(4))
dy/dx w.r.t. : yr89 male white age ed prst

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   .0729985   .0111797     6.53   0.000     .0510866    .0949103
        male |   -.102175   .0111936    -9.13   0.000    -.1241141    -.080236
       white |  -.0545026    .016485    -3.31   0.001    -.0868126   -.0221926
         age |  -.0030188   .0003501    -8.62   0.000    -.0037051   -.0023325
          ed |   .0093596   .0022406     4.18   0.000     .0049681    .0137511
        prst |   .0008461   .0004584     1.85   0.065    -.0000522    .0017445
------------------------------------------------------------------------------
```

3.4.3 Marginal Effects at the Average

```
ologit warm yr89 male white age ed prst, or

margins, dydx(*) atmeans predict(outcome(1))
margins, dydx(*) atmeans predict(outcome(2))
margins, dydx(*) atmeans predict(outcome(3))
margins, dydx(*) atmeans predict(outcome(4))
```

```
Iteration 0:   log likelihood = -2995.7704
Iteration 1:   log likelihood = -2846.4532
Iteration 2:   log likelihood = -2844.9142
Iteration 3:   log likelihood = -2844.9123
Iteration 4:   log likelihood = -2844.9123

Ordered logistic regression                     Number of obs     =      2,293
                                                LR chi2(6)        =     301.72
                                                Prob > chi2       =     0.0000
Log likelihood = -2844.9123                     Pseudo R2         =     0.0504

------------------------------------------------------------------------------
        warm | Odds Ratio   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   1.688605   .1349176     6.56   0.000     1.443836    1.974868
        male |   .4803214   .0376969    -9.34   0.000     .4118389    .5601916
       white |   .6762723   .0800577    -3.30   0.001     .5362356    .8528792
         age |   .9785675   .0024154    -8.78   0.000     .9738449     .983313
          ed |    1.06948   .0170849     4.20   0.000     1.036513    1.103496
        prst |   1.006091    .003313     1.84   0.065     .9996188    1.012605
-------------+----------------------------------------------------------------
       /cut1 |  -2.465362   .2389128                     -2.933622   -1.997102
       /cut2 |   -.630904   .2333156                     -1.088194   -.1736138
       /cut3 |   1.261854    .234018                      .8031871    1.720521
------------------------------------------------------------------------------


Conditional marginal effects                    Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==1), predict(outcome(1))
dy/dx w.r.t. : yr89 male white age ed prst
at           : yr89            =    .3986044 (mean)
               male            =    .4648932 (mean)
               white           =    .8765809 (mean)
               age             =    44.93546 (mean)
               ed              =    12.21805 (mean)
               prst            =    39.58526 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   -.051803   .0081383    -6.37   0.000    -.0677537   -.0358522
        male |   .0725079   .0081294     8.92   0.000     .0565746    .0884413
       white |   .0386775   .0118016     3.28   0.001     .0155468    .0618081
```

```
         age |   .0021423   .0002546     8.41   0.000     .0016432    .0026413
          ed |   -.006642   .0015972    -4.16   0.000    -.0097724   -.0035116
        prst |  -.0006005   .0003264    -1.84   0.066    -.0012401    .0000392
------------------------------------------------------------------------------


Conditional marginal effects                    Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==2), predict(outcome(2))
dy/dx w.r.t. : yr89 male white age ed prst
at           : yr89            =    .3986044 (mean)
               male            =    .4648932 (mean)
               white           =    .8765809 (mean)
               age             =    44.93546 (mean)
               ed              =    12.21805 (mean)
               prst            =    39.58526 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |  -.0772501   .0122256    -6.32   0.000    -.1012118   -.0532885
        male |   .1081261   .0124796     8.66   0.000     .0836666    .1325855
       white |    .057677    .017613     3.27   0.001     .0231561    .0921979
         age |   .0031946   .0003899     8.19   0.000     .0024304    .0039588
          ed |  -.0099047   .0023955    -4.13   0.000    -.0145999   -.0052095
        prst |  -.0008954   .0004869    -1.84   0.066    -.0018497    .0000588
------------------------------------------------------------------------------


Conditional marginal effects                    Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==3), predict(outcome(3))
dy/dx w.r.t. : yr89 male white age ed prst
at           : yr89            =    .3986044 (mean)
               male            =    .4648932 (mean)
               white           =    .8765809 (mean)
               age             =    44.93546 (mean)
               ed              =    12.21805 (mean)
               prst            =    39.58526 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
```

```
-------------+----------------------------------------------------------------
        yr89 |   .0582103   .0096161     6.05   0.000     .0393631    .0770575
        male |  -.0814762   .0099979    -8.15   0.000    -.1010717   -.0618807
       white |  -.0434613   .0134313    -3.24   0.001    -.0697862   -.0171364
         age |  -.0024072   .0003126    -7.70   0.000    -.0030199   -.0017945
          ed |   .0074635   .0018362     4.06   0.000     .0038645    .0110625
        prst |   .0006747   .0003683     1.83   0.067    -.0000472    .0013967
------------------------------------------------------------------------------


Conditional marginal effects                    Number of obs     =      2,293
Model VCE    : OIM

Expression   : Pr(warm==4), predict(outcome(4))
dy/dx w.r.t. : yr89 male white age ed prst
at           : yr89            =    .3986044 (mean)
               male            =    .4648932 (mean)
               white           =    .8765809 (mean)
               age             =    44.93546 (mean)
               ed              =    12.21805 (mean)
               prst            =    39.58526 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yr89 |   .0708428   .0109038     6.50   0.000     .0494718    .0922139
        male |  -.0991578   .0109073    -9.09   0.000    -.1205356   -.0777799
       white |  -.0528931   .0160474    -3.30   0.001    -.0843454   -.0214409
         age |  -.0029296     .00034    -8.62   0.000    -.0035961   -.0022632
          ed |   .0090832   .0021702     4.19   0.000     .0048296    .0133368
        prst |   .0008212   .0004455     1.84   0.065    -.0000519    .0016943
------------------------------------------------------------------------------
```

Chapter 4

Reshape

From my experience, I feel that understanding and mastering the reshape command is invaluable. This is especially true when working with panel data. Panel data should not be used in a wide format. Panel data should be in a long format where you can use the xtset unit time command. After the xtset is established you can easily add lags or differences with the l.var or d.var

It is also helpful in terms of merging datasets. A lot of time if I'm merging BLS data, such as state unemployment, or inflation rates, I need to reshape the BLS data from wide to long to I can merge m:1

I typically work in long formats, but reshaping to wide formats does have its benefits. I find wide formats are helpful for people who work in excel, especially when years are across columns.

4.1 Wide and Long Datasets

We have two types of datasets: wide and long

Wide Format

Our wide datasets typically include years or variables types spread out across the columns.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
Contains data from cardio_wide.dta
  obs:             6
 vars:            12                          22 Dec 2009 20:43
 size:           120
--------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
--------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
age             byte    %3.0f                 Age of person
bp1             int     %3.0f                 Blood pressure systolic Trial 1
bp2             int     %3.0f                 Blood pressure systolic Trial 2
bp3             int     %3.0f                 Blood pressure systolic Trial 3
bp4             int     %3.0f                 Blood pressure systolic Trial 4
bp5             int     %3.0f                 Blood pressure systolic Trial 5
pl1             int     %3.0f                 Pulse: Trial 1
pl2             byte    %3.0f                 Pulse: Trial 2
pl3             int     %3.0f                 Pulse: Trial 3
pl4             int     %3.0f                 Pulse: Trial 4
pl5             byte    %3.0f                 Pulse: Trial 5
--------------------------------------------------------------------------------
Sorted by:
```

We have 12 variables idcode, age, and our 5 different blood pressure and pulse trials. We see that are 5 different incidents/trials of blood pressure and pulse are spread out across 5 different variables $bp1 - bp5$ and $pl1 - pl5$*

```
list
```

```
     | id   age   bp1   bp2   bp3   bp4   bp5   pl1   pl2   pl3   pl4   pl5 |
     |----------------------------------------------------------------------|
  1. |  1    40   115    86   129   105   127    54    87    93    81    92 |
  2. |  2    30   123   136   107   111   120    92    88   125    87    58 |
  3. |  3    16   124   122   101   109   112   105    97   128    57    68 |
  4. |  4    23   105   115   121   129   137    52    79    71   106    39 |
  5. |  5    18   116   128   112   125   111    70    64    52    68    59 |
     |----------------------------------------------------------------------|
  6. |  6    27   108   126   124   131   107    74    78    92    99    80 |
     +----------------------------------------------------------------------+
```

With a wide format each row is an observation and all of our different trials are located on a single row

Long Format

We'll look at an example of the same dataset but in long format

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



Contains data from cardio_long.dta
  obs:            30
 vars:             5                          10 Feb 2020 23:02
 size:           210
----------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
----------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trial           byte    %9.0g                 Trial number
age             byte    %3.0f                 Age of person
bp              int     %3.0f                 Blood pressure (systolic)
pl              int     %3.0f                 Pulse
----------------------------------------------------------------------------------------------
Sorted by: id  trial
```

We only have 5 variables idcode, age, trial, blood pressure, and pulse.

```
list
```

```
     | id   trial   age    bp    pl |
     |------------------------------|
  1. |  1       1    40   115    54 |
  2. |  1       2    40    86    87 |
  3. |  1       3    40   129    93 |
  4. |  1       4    40   105    81 |
  5. |  1       5    40   127    92 |
     |------------------------------|
  6. |  2       1    30   123    92 |
  7. |  2       2    30   136    88 |
  8. |  2       3    30   107   125 |
  9. |  2       4    30   111    87 |
 10. |  2       5    30   120    58 |
```

```
     |-----------------------------|
 11. |  3       1    16    124   105 |
 12. |  3       2    16    122    97 |
 13. |  3       3    16    101   128 |
 14. |  3       4    16    109    57 |
 15. |  3       5    16    112    68 |
     |-----------------------------|
 16. |  4       1    23    105    52 |
 17. |  4       2    23    115    79 |
 18. |  4       3    23    121    71 |
 19. |  4       4    23    129   106 |
 20. |  4       5    23    137    39 |
     |-----------------------------|
 21. |  5       1    18    116    70 |
 22. |  5       2    18    128    64 |
 23. |  5       3    18    112    52 |
 24. |  5       4    18    125    68 |
 25. |  5       5    18    111    59 |
     |-----------------------------|
 26. |  6       1    27    108    74 |
 27. |  6       2    27    126    78 |
 28. |  6       3    27    124    92 |
 29. |  6       4    27    131    99 |
 30. |  6       5    27    107    80 |
     +-----------------------------+
```

We also see that each individual has 5 observations. One for each trial that is sorted from 1 to 5.

Notice: We have a cross-sectional unit variable with *idcode* and we also have a time component with *trial*. Our dataset is basically a panel data set. We observe the same person over 5 trials.

What kind of format should we use?

It depends.

Wide Format

Mitchell makes a good point of looking at correlations among the blood pressure trials.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
correlate bp1 bp2 bp3 bp4 bp5
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell


(obs=6)

             |      bp1      bp2      bp3      bp4      bp5
-------------+---------------------------------------------
         bp1 |   1.0000
         bp2 |   0.2427   1.0000
         bp3 |  -0.7662  -0.6657   1.0000
         bp4 |  -0.7644   0.3980   0.2644   1.0000
         bp5 |  -0.3643  -0.4984   0.3694  -0.0966   1.0000
```

Mitchell also mentions that multivariate analysis with multiple trials is a bit easier with factor variables. If we wanted to see all of the different regressions for each trial, then

```
mvreg bp* = age
```

```
Equation             Obs   Parms        RMSE    "R-sq"          F        P
--------------------------------------------------------------------------
bp1                    6       2    8.516293    0.0160   .0651388   0.8111
bp2                    6       2    15.34222    0.3858   2.512064   0.1882
bp3                    6       2    8.846404    0.4597   3.402782   0.1388
bp4                    6       2    11.60081    0.1554   .7357661   0.4394
bp5                    6       2    11.81248    0.1360    .629676   0.4719

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
bp1          |
         age |  -.1107266   .4338427    -0.26   0.811    -1.315267    1.093814
       _cons |   118.0087   11.66545    10.12   0.001     85.62017    150.3971
-------------+----------------------------------------------------------------
bp2          |
         age |  -1.238754   .7815736    -1.58   0.188     -3.40875    .9312418
       _cons |    150.628   21.01547     7.17   0.002     92.27974    208.9763
-------------+----------------------------------------------------------------
bp3          |
         age |   .8313149   .4506595     1.84   0.139    -.4199164    2.082546
       _cons |   94.32958   12.11763     7.78   0.001     60.68565    127.9735
-------------+----------------------------------------------------------------
bp4          |
         age |  -.5069204   .5909761    -0.86   0.439    -2.147733    1.133892
       _cons |   131.3443   15.89055     8.27   0.001     87.22504    175.4635
```

```
-------------+----------------------------------------------------------------
bp5          |
         age |   .4775087   .6017591     0.79   0.472    -1.193242     2.14826
       _cons |   106.7439    16.1805     6.60   0.003     61.81969    151.6682
------------------------------------------------------------------------------
```

Long Format

There are several reasons we would want to use long format over wide format

Correlations

If we want correlations between blood pressure and pulse, a long format is preferable.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
correlate bp pl
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell


(obs=30)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |  -0.0444   1.0000
```

Or

```
bysort trial: correlate bp pl
```

```
-> trial = 1
(obs=6)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |   0.7958   1.0000


-----------------------------------------------------------------------------------------------
```

```
-> trial = 2
(obs=6)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |  -0.1985   1.0000


-----------------------------------------------------------------------------------------------
-> trial = 3
(obs=6)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |  -0.4790   1.0000


-----------------------------------------------------------------------------------------------
-> trial = 4
(obs=6)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |   0.5639   1.0000


-----------------------------------------------------------------------------------------------
-> trial = 5
(obs=6)

             |       bp       pl
-------------+------------------
          bp |   1.0000
          pl |  -0.3911   1.0000
```

Panel data

It is also more appropriate for panel data and the xt commands, such as *xtset*, *xtreg*, etc. I also prefer this for regular regressions where we can use factor variables *i.var* for different time periods.

```
xtset id trial
xtreg bp age
xtline bp, overlay
```

Recoding

It is also easier to recode in long format.

```
recode pl (min/89=0) (90/max=1), generate (plhi)
```

Recoding among many variables in wide format is more of a pain You can see this on page 293 in Mitchell.

Egen is easier

If you wanted to mean of each 5 trials, it just as easy with egen in long format.

```
bysort id: egen pl_avg=mean(pl)
bysort id: egen bp_avg=mean(bp)
```

For wide-format you can use rowmean with the wildcard

```
use cardio_wide, clear
```

We don't need the bysort group, but we need to be careful which columns to include

```
egen pl_avg = rowmean(pl*)
egen bp_avg = rowmean(bp*)
```

It easier to explicitly difference trails with wide-format, but requires more typing.

```
generate pldiff2 = pl2-pl1
generate pldiff3 = pl3-pl2
generate pldiff4 = pl4-pl3
generate pldiff5 = pl5-pl4
```

4.1.0.1 Using subscripting/indexing, differences, lags with ease

I prefer using subscripting/indexing within groups or using xtset with panel data since we have a lot of flexibilities.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
sort id trial
```

We can use indexing

```
by id: generate pldiff = pl-pl[_n-1]
```

We can difference from the first trial

```
by id: generate pldiff_1trial = pl-pl[1]
```

We can use xtset as well with l.var

```
xtset id trial
```

```
       panel variable:  id (strongly balanced)
        time variable:  trial, 1 to 5
                delta:  1 unit
```

One lag

```
gen pldiff_other = pl-l.pl
```

We can use two lags

```
gen pldiff2_other = l2.pl-l1.pl
```

We can use three lags

```
gen pldiff3_other = l3.pl-l2.pl
list
```

```
(18 missing values generated)

     +------------------------------------------------------------------------------------+
     | id   trial   age    bp    pl   pldiff   pldiff~l   ~f_other   p~2_ot~r   p~3_ot~r |
```

```
     |---------------------------------------------------------------------------
  1. | 1        1   40   115    54         .          0          .          .
  2. | 1        2   40    86    87        33         33         33          .
  3. | 1        3   40   129    93         6         39          6        -33
  4. | 1        4   40   105    81       -12         27        -12         -6        -3
  5. | 1        5   40   127    92        11         38         11         12        -
     |---------------------------------------------------------------------------
  6. | 2        1   30   123    92         .          0          .          .
  7. | 2        2   30   136    88        -4         -4         -4          .
  8. | 2        3   30   107   125        37         33         37          4
  9. | 2        4   30   111    87       -38         -5        -38        -37
 10. | 2        5   30   120    58       -29        -34        -29         38        -3
     |---------------------------------------------------------------------------
 11. | 3        1   16   124   105         .          0          .          .
 12. | 3        2   16   122    97        -8         -8         -8          .
 13. | 3        3   16   101   128        31         23         31          8
 14. | 3        4   16   109    57       -71        -48        -71        -31
 15. | 3        5   16   112    68        11        -37         11         71        -3
     |---------------------------------------------------------------------------
 16. | 4        1   23   105    52         .          0          .          .
 17. | 4        2   23   115    79        27         27         27          .
 18. | 4        3   23   121    71        -8         19         -8        -27
 19. | 4        4   23   129   106        35         54         35          8        -2
 20. | 4        5   23   137    39       -67        -13        -67        -35
     |---------------------------------------------------------------------------
 21. | 5        1   18   116    70         .          0          .          .
 22. | 5        2   18   128    64        -6         -6         -6          .
 23. | 5        3   18   112    52       -12        -18        -12          6
 24. | 5        4   18   125    68        16         -2         16         12
 25. | 5        5   18   111    59        -9        -11         -9        -16         1
     |---------------------------------------------------------------------------
 26. | 6        1   27   108    74         .          0          .          .
 27. | 6        2   27   126    78         4          4          4          .
 28. | 6        3   27   124    92        14         18         14         -4
 29. | 6        4   27   131    99         7         25          7        -14         -
 30. | 6        5   27   107    80       -19          6        -19         -7        -1
     +---------------------------------------------------------------------------
```

Wide format can become unwieldy quickly and Stata does have a variable limit size, while your observation limit is typically your memory.In short, it is better in the long-run to learn to work with Stata in long-format.

However, Mitchell does bring up a good point that in our panel data, we mayneed multiple weights wt1-wtk. You can see this in the ACS PUMS with 250 or soreplicate weights. We do not want to reshape our data by weights, we need to keep the data in a cross-sectional unit i and time period t format.

In short, I agree with Mitchell that you should learn to work with your data in a long-format data structure.

4.2 Reshaping Long to Wide

There may be scenarios where we need to reshape our long-format data into a wide format. For example, it might be easier to match and merge our data If our using data set is in wide format and our master data set is in wide format

Let's look at an example

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
describe
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



Contains data from cardio_long.dta
  obs:            30
 vars:             5                          10 Feb 2020 23:02
 size:           210
------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trial           byte    %9.0g                 Trial number
age             byte    %3.0f                 Age of person
bp              int     %3.0f                 Blood pressure (systolic)
pl              int     %3.0f                 Pulse
------------------------------------------------------------------------------------------
Sorted by: id  trial

     +------------------------------+
     | id   trial   age    bp    pl |
     |------------------------------|
  1. |  1       1    40   115    54 |
  2. |  1       2    40    86    87 |
  3. |  1       3    40   129    93 |
  4. |  1       4    40   105    81 |
  5. |  1       5    40   127    92 |
```

```
     |-----------------------------|
  6. |  2       1    30   123    92 |
  7. |  2       2    30   136    88 |
  8. |  2       3    30   107   125 |
  9. |  2       4    30   111    87 |
 10. |  2       5    30   120    58 |
     |-----------------------------|
 11. |  3       1    16   124   105 |
 12. |  3       2    16   122    97 |
 13. |  3       3    16   101   128 |
 14. |  3       4    16   109    57 |
 15. |  3       5    16   112    68 |
     |-----------------------------|
 16. |  4       1    23   105    52 |
 17. |  4       2    23   115    79 |
 18. |  4       3    23   121    71 |
 19. |  4       4    23   129   106 |
 20. |  4       5    23   137    39 |
     |-----------------------------|
 21. |  5       1    18   116    70 |
 22. |  5       2    18   128    64 |
 23. |  5       3    18   112    52 |
 24. |  5       4    18   125    68 |
 25. |  5       5    18   111    59 |
     |-----------------------------|
 26. |  6       1    27   108    74 |
 27. |  6       2    27   126    78 |
 28. |  6       3    27   124    92 |
 29. |  6       4    27   131    99 |
 30. |  6       5    27   107    80 |
     +-----------------------------+
```

Our reshape command requires *reshape* 1. What direction: wide or long 2. Variables to reshape (bp and pl) 3. Variables define an observation i(id) 4. Variables that defines the repeated observations for each person orvtime variable j(trial) for reshaping long to wide

Trial values will become the suffix values for our reshape values

```
reshape wide bp pl, i(id) j(trial)
describe
list
```

```
(note: j = 1 2 3 4 5)

Data                                   long   ->   wide
```

```
----------------------------------------------------------------------------
Number of obs.                         30   ->       6
Number of variables                     5   ->      12
j variable (5 values)               trial   ->   (dropped)
xij variables:
                                       bp   ->   bp1 bp2 ... bp5
                                       pl   ->   pl1 pl2 ... pl5
----------------------------------------------------------------------------


Contains data
  obs:             6
 vars:            12
 size:           132
---------------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
---------------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
bp1             int     %3.0f                 1 bp
pl1             int     %3.0f                 1 pl
bp2             int     %3.0f                 2 bp
pl2             int     %3.0f                 2 pl
bp3             int     %3.0f                 3 bp
pl3             int     %3.0f                 3 pl
bp4             int     %3.0f                 4 bp
pl4             int     %3.0f                 4 pl
bp5             int     %3.0f                 5 bp
pl5             int     %3.0f                 5 pl
age             byte    %3.0f                 Age of person
---------------------------------------------------------------------------------------------------
Sorted by: id
     Note: Dataset has changed since last saved.

     +-----------------------------------------------------------------------+
     | id   bp1   pl1   bp2   pl2   bp3   pl3   bp4   pl4   bp5   pl5   age |
     |-----------------------------------------------------------------------|
  1. |  1   115    54    86    87   129    93   105    81   127    92    40 |
  2. |  2   123    92   136    88   107   125   111    87   120    58    30 |
  3. |  3   124   105   122    97   101   128   109    57   112    68    16 |
  4. |  4   105    52   115    79   121    71   129   106   137    39    23 |
  5. |  5   116    70   128    64   112    52   125    68   111    59    18 |
     |-----------------------------------------------------------------------|
  6. |  6   108    74   126    78   124    92   131    99   107    80    27 |
     +-----------------------------------------------------------------------+
```

Notice we excluded age. It is constant within the cross-sectional unit. It does not define the cross-sectional unit or the*time dimension unit. We do not want to separate it into repeated observations either.

Without doing any additional work, we can reshape it back to long with simply reshape long.

```
reshape long
describe
```

```
(note: j = 1 2 3 4 5)

Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      30
Number of variables                  12   ->       5
j variable (5 values)                     ->   trial
xij variables:
                        bp1 bp2 ... bp5   ->   bp
                        pl1 pl2 ... pl5   ->   pl
-----------------------------------------------------------------------------
```

```
Contains data
  obs:            30
 vars:             5
 size:           210
---------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
---------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trial           byte    %9.0g                 Trial number
bp              int     %3.0f                 Blood pressure (systolic)
pl              int     %3.0f                 Pulse
age             byte    %3.0f                 Age of person
---------------------------------------------------------------------------------
Sorted by: id  trial
     Note: Dataset has changed since last saved.
```

4.3 Reshaping Long to Wide: Problems

Problem 1: A constant value within a cross-sectional unit is not constant

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long2, clear
list in 1/10
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell

     +------------------------------+
     | id   trial   age    bp    pl |
     |------------------------------|
  1. |  1       1    40   115    54 |
  2. |  1       2    40    86    87 |
  3. |  1       3    44   129    93 |
  4. |  1       4    40   105    81 |
  5. |  1       5    40   127    92 |
     |------------------------------|
  6. |  2       1    30   123    92 |
  7. |  2       2    30   136    88 |
  8. |  2       3    30   107   125 |
  9. |  2       4    30   111    87 |
 10. |  2       5    30   120    58 |
     +------------------------------+
```

We see that cross-sectional unit 1 has a data entry problem with age in the 3rd trial. If we reshape, we will get an error

```
reshape wide bp pl, i(id) j(trial)
```

```
(note: j = 1 2 3 4 5)
variable age not constant within id
    Your data are currently long.  You are performing a reshape wide.  You typed something
    like

        . reshape wide a b, i(id) j(trial)

    There are variables other than a, b, id, trial in your data.  They must be constant
    within id because that is the only way they can fit into wide data without loss of
    information.
```

```
    The variable or variables listed above are not constant within id.  Perhaps the val
    are in error.  Type reshape error for a list of the problem observations.

    Either that, or the values vary because they should vary, in which case you must ei
    add the variables to the list of xij variables to be reshaped, or drop them.
r(9);

r(9);
```

Use the reshape error command to help with the problem

```
reshape error
```

It tells us that our constant variable is not constant within cross-sectional units.

Problem 2: Excluding a non-constant that we need to reshape

If we excluded pl from our reshape, we will get a similar error.

```
capture reshape wide bp, i(id) j(trial)
reshape error
```

```
(note: j = 1 2 3 4 5)

i (id) indicates the top-level grouping such as subject id.
j (trial) indicates the subgrouping such as time.
xij variable is bp.
Thus, the following variable(s) should be constant within i:
      age pl

age not constant within i (id) for 1 value of i:

     +------------------+
     | id   trial   age |
     |------------------|
  1. |  1       1    40 |
  2. |  1       2    40 |
  3. |  1       3    44 |
  4. |  1       4    40 |
  5. |  1       5    40 |
     +------------------+
```

```
pl not constant within i (id) for 6 values of i:

     +------------------+
     | id   trial    pl |
     |------------------|
  1. |  1       1    54 |
  2. |  1       2    87 |
  3. |  1       3    93 |
  4. |  1       4    81 |
  5. |  1       5    92 |
     |------------------|
  6. |  2       1    92 |
  7. |  2       2    88 |
  8. |  2       3   125 |
  9. |  2       4    87 |
 10. |  2       5    58 |
     |------------------|
 11. |  3       1   105 |
 12. |  3       2    97 |
 13. |  3       3   128 |
 14. |  3       4    57 |
 15. |  3       5    68 |
     |------------------|
 16. |  4       1    52 |
 17. |  4       2    79 |
 18. |  4       3    71 |
 19. |  4       4   106 |
 20. |  4       5    39 |
     |------------------|
 21. |  5       1    70 |
 22. |  5       2    64 |
 23. |  5       3    52 |
 24. |  5       4    68 |
 25. |  5       5    59 |
     |------------------|
 26. |  6       1    74 |
 27. |  6       2    78 |
 28. |  6       3    92 |
 29. |  6       4    99 |
 30. |  6       5    80 |
     +------------------+

(data now sorted by id trial)
```

4.4 Reshaping Wide to Long

In my personal opinion, this reshaping wide to long is a more common occurance. It is important to get data into a long-format so we can conduct panel analysis. I had to do this for a few of the Wooldridge datasets that were panel data, but in wide format.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
describe
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



Contains data from cardio_wide.dta
  obs:             6
 vars:            12                          22 Dec 2009 20:43
 size:           120
--------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
--------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
age             byte    %3.0f                 Age of person
bp1             int     %3.0f                 Blood pressure systolic Trial 1
bp2             int     %3.0f                 Blood pressure systolic Trial 2
bp3             int     %3.0f                 Blood pressure systolic Trial 3
bp4             int     %3.0f                 Blood pressure systolic Trial 4
bp5             int     %3.0f                 Blood pressure systolic Trial 5
pl1             int     %3.0f                 Pulse: Trial 1
pl2             byte    %3.0f                 Pulse: Trial 2
pl3             int     %3.0f                 Pulse: Trial 3
pl4             int     %3.0f                 Pulse: Trial 4
pl5             byte    %3.0f                 Pulse: Trial 5
--------------------------------------------------------------------------------------
Sorted by:

     +----------------------------------------------------------------------+
     | id   age   bp1   bp2   bp3   bp4   bp5   pl1   pl2   pl3   pl4   pl5 |
     |----------------------------------------------------------------------|
  1. |  1    40   115    86   129   105   127    54    87    93    81    92 |
  2. |  2    30   123   136   107   111   120    92    88   125    87    58 |
  3. |  3    16   124   122   101   109   112   105    97   128    57    68 |
```

```
  4. |  4    23   105   115   121   129   137    52    79    71   106    39 |
  5. |  5    18   116   128   112   125   111    70    64    52    68    59 |
     |-----------------------------------------------------------------------|
  6. |  6    27   108   126   124   131   107    74    78    92    99    80 |
     +-----------------------------------------------------------------------+
```

Let's reshape wide to long.

Our reshape values that are not constant: bp and pl.

Our cross-sectional unit to reshape upon or our list of variables that uniquely identify a cross-sectional unit. For example, we may need a State FIPS code and a County FIPS code to unique identify a county cross-sectional unit (of course we can merge the 2-digit State FIPS code and the 3-digit County FIPS code to create a 5-digit FIPS code in one variable).

Our time unit, where with reshape long, we need to create a new variable that takes on the values of the suffixes of bp and pl (For example trialnum). Our constant values within cross-sectional units do not need to be identified.

```
reshape long bp pl, i(id) j(trialnum)
describe
list in 1/10
```

```
(note: j = 1 2 3 4 5)

Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      30
Number of variables                  12   ->       5
j variable (5 values)                     ->   trialnum
xij variables:
                        bp1 bp2 ... bp5   ->   bp
                        pl1 pl2 ... pl5   ->   pl
-----------------------------------------------------------------------------


Contains data
  obs:            30
 vars:             5
 size:           210
----------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
----------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trialnum        byte    %9.0g
```

```
age             byte    %3.0f                Age of person
bp              int     %3.0f
pl              int     %3.0f
-----------------------------------------------------------------------------------------
Sorted by: id  trialnum
     Note: Dataset has changed since last saved.

     +----------------------------------+
     | id   trialnum   age    bp     pl |
     |----------------------------------|
  1. |  1          1    40   115     54 |
  2. |  1          2    40    86     87 |
  3. |  1          3    40   129     93 |
  4. |  1          4    40   105     81 |
  5. |  1          5    40   127     92 |
     |----------------------------------|
  6. |  2          1    30   123     92 |
  7. |  2          2    30   136     88 |
  8. |  2          3    30   107    125 |
  9. |  2          4    30   111     87 |
 10. |  2          5    30   120     58 |
     +----------------------------------+
```

To reshape it back to wide, we just simply state

```
reshape wide
describe
list
```

```
(note: j = 1 2 3 4 5)

Data                               long   ->   wide
-----------------------------------------------------------------------------
Number of obs.                       30   ->       6
Number of variables                   5   ->      12
j variable (5 values)          trialnum   ->   (dropped)
xij variables:
                                     bp   ->   bp1 bp2 ... bp5
                                     pl   ->   pl1 pl2 ... pl5
-----------------------------------------------------------------------------


Contains data
  obs:             6
 vars:            12
```

```
 size:           132
-----------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
-----------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
bp1             int     %3.0f                 1 bp
pl1             int     %3.0f                 1 pl
bp2             int     %3.0f                 2 bp
pl2             int     %3.0f                 2 pl
bp3             int     %3.0f                 3 bp
pl3             int     %3.0f                 3 pl
bp4             int     %3.0f                 4 bp
pl4             int     %3.0f                 4 pl
bp5             int     %3.0f                 5 bp
pl5             int     %3.0f                 5 pl
age             byte    %3.0f                 Age of person
-----------------------------------------------------------------------------------------------
Sorted by: id
     Note: Dataset has changed since last saved.

     +-----------------------------------------------------------------------+
     | id   bp1   pl1   bp2   pl2   bp3   pl3   bp4   pl4   bp5   pl5   age |
     |-----------------------------------------------------------------------|
  1. |  1   115    54    86    87   129    93   105    81   127    92    40 |
  2. |  2   123    92   136    88   107   125   111    87   120    58    30 |
  3. |  3   124   105   122    97   101   128   109    57   112    68    16 |
  4. |  4   105    52   115    79   121    71   129   106   137    39    23 |
  5. |  5   116    70   128    64   112    52   125    68   111    59    18 |
     |-----------------------------------------------------------------------|
  6. |  6   108    74   126    78   124    92   131    99   107    80    27 |
     +-----------------------------------------------------------------------+
```

4.5 Reshaping wide to long problems

Problem 1: Failing to list all of the varying variables to be reshaped.

This is a dangerous kind of problem, since it doesn't throw an error. This requires the user to inspect the data after reshaping to make sure everything worked properly.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide, clear
reshape long bp, i(id) j(trialnum)
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell


(note: j = 1 2 3 4 5)

Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      30
Number of variables                  12   ->       9
j variable (5 values)                     ->   trialnum
xij variables:
                        bp1 bp2 ... bp5   ->   bp
-----------------------------------------------------------------------------


     +----------------------------------------------------------+
     | id   trialnum   age    bp   pl1   pl2   pl3   pl4   pl5 |
     |----------------------------------------------------------|
  1. |  1          1    40   115    54    87    93    81    92 |
  2. |  1          2    40    86    54    87    93    81    92 |
  3. |  1          3    40   129    54    87    93    81    92 |
  4. |  1          4    40   105    54    87    93    81    92 |
  5. |  1          5    40   127    54    87    93    81    92 |
     |----------------------------------------------------------|
  6. |  2          1    30   123    92    88   125    87    58 |
  7. |  2          2    30   136    92    88   125    87    58 |
  8. |  2          3    30   107    92    88   125    87    58 |
  9. |  2          4    30   111    92    88   125    87    58 |
 10. |  2          5    30   120    92    88   125    87    58 |
     |----------------------------------------------------------|
 11. |  3          1    16   124   105    97   128    57    68 |
 12. |  3          2    16   122   105    97   128    57    68 |
 13. |  3          3    16   101   105    97   128    57    68 |
 14. |  3          4    16   109   105    97   128    57    68 |
 15. |  3          5    16   112   105    97   128    57    68 |
     |----------------------------------------------------------|
 16. |  4          1    23   105    52    79    71   106    39 |
 17. |  4          2    23   115    52    79    71   106    39 |
 18. |  4          3    23   121    52    79    71   106    39 |
 19. |  4          4    23   129    52    79    71   106    39 |
 20. |  4          5    23   137    52    79    71   106    39 |
```

```
     |---------------------------------------------------------|
 21. |  5          1    18   116     70     64     52     68     59 |
 22. |  5          2    18   128     70     64     52     68     59 |
 23. |  5          3    18   112     70     64     52     68     59 |
 24. |  5          4    18   125     70     64     52     68     59 |
 25. |  5          5    18   111     70     64     52     68     59 |
     |---------------------------------------------------------|
 26. |  6          1    27   108     74     78     92     99     80 |
 27. |  6          2    27   126     74     78     92     99     80 |
 28. |  6          3    27   124     74     78     92     99     80 |
 29. |  6          4    27   131     74     78     92     99     80 |
 30. |  6          5    27   107     74     78     92     99     80 |
     +---------------------------------------------------------+
```

The remedy is easy: don't forget all of your variables to be reshaped

```
use cardio_wide, clear
reshape long bp pl, i(id) j(trialnum)
```

Problem 2: Be careful when the time-varying value is embedded within the variable name and not at the end.

Typically you will find that the varying variable will have $var_1, var_2, ..., var_k$ and you simply specify reshape long var, i(id) j(time). But, if the varying value is embedded like t1bp,...,t5bp and t1pl,...,t5pl, then we cannot simply use reshape long tbp tpl, i(id) j(trialnum), since Stata looks for the varying value at the end.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide2, clear
describe
capture reshape long tpl tpb, i(id) j(trialnum)
reshape error
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



Contains data from cardio_wide2.dta
  obs:             6
 vars:            12                        31 Dec 2009 15:46
 size:           120
--------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
```

```
--------------------------------------------------------------------------------
id              byte    %3.0f                ID of person
age             byte    %3.0f                Age of person
t1bp            int     %3.0f                Blood pressure systolic Trial 1
t2bp            int     %3.0f                Blood pressure systolic Trial 2
t3bp            int     %3.0f                Blood pressure systolic Trial 3
t4bp            int     %3.0f                Blood pressure systolic Trial 4
t5bp            int     %3.0f                Blood pressure systolic Trial 5
t1pl            int     %3.0f                Pulse: Trial 1
t2pl            byte    %3.0f                Pulse: Trial 2
t3pl            int     %3.0f                Pulse: Trial 3
t4pl            int     %3.0f                Pulse: Trial 4
t5pl            byte    %3.0f                Pulse: Trial 5
--------------------------------------------------------------------------------
Sorted by:


no xij variables found
    You typed something like reshape wide a b, i(i) j(j).
    reshape looked for existing variables named a# and b# but could not find any.  Reme
    this picture:

         long                                 wide
        +---------------+                    +------------------+
        | i   j   a   b |                    | i   a1 a2  b1 b2 |
        |---------------| <--- reshape --->  |------------------|
        | 1   1   1   2 |                    | 1   1   3   2  4 |
        | 1   2   3   4 |                    | 2   5   7   6  8 |
        | 2   1   5   6 |                    +------------------+
        | 2   2   7   8 |
        +---------------+

        long to wide: reshape wide a b, i(i) j(j)    (j existing variable)
        wide to long: reshape long a b, i(i) j(j)    (j    new   variable)
```

We need the operator @ to indicate to Stata that our varying-values are in that part of the variable names.

```
reshape long t@pl t@bp, i(id) j(trialnum)
describe
list in 1/10
```

```
(note: j = 1 2 3 4 5)

Data                                   wide   ->   long
```

```
-----------------------------------------------------------------------------
Number of obs.                         6   ->      30
Number of variables                   12   ->       5
j variable (5 values)                      ->   trialnum
xij variables:
                     t1pl t2pl ... t5pl   ->   tpl
                     t1bp t2bp ... t5bp   ->   tbp
-----------------------------------------------------------------------------


Contains data
  obs:            30
 vars:             5
 size:           210
--------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
--------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trialnum        byte    %9.0g
age             byte    %3.0f                 Age of person
tbp             int     %3.0f
tpl             int     %3.0f
--------------------------------------------------------------------------------------------
Sorted by: id  trialnum
     Note: Dataset has changed since last saved.

     +---------------------------------+
     | id   trialnum   age   tbp   tpl |
     |---------------------------------|
  1. |  1          1    40   115    54 |
  2. |  1          2    40    86    87 |
  3. |  1          3    40   129    93 |
  4. |  1          4    40   105    81 |
  5. |  1          5    40   127    92 |
     |---------------------------------|
  6. |  2          1    30   123    92 |
  7. |  2          2    30   136    88 |
  8. |  2          3    30   107   125 |
  9. |  2          4    30   111    87 |
 10. |  2          5    30   120    58 |
     +---------------------------------+
```

Problem 3: Be specific with which variables to reshape and don't mistakenly constant variables that have the same prefix as the varying variables

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
clist bp* pl*, noobs
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



bp1  bp2  bp3  bp4  bp5     bp2005  pl1  pl2  pl3  pl4  pl5     pl2005
115   86  129  105  127        112   54   87   93   81   92         81
123  136  107  111  120        119   92   88  125   87   58         90
124  122  101  109  112        113  105   97  128   57   68         91
105  115  121  129  137        121   52   79   71  106   39         69
116  128  112  125  111        118   70   64   52   68   59         62
108  126  124  131  107        119   74   78   92   99   80         84
```

Our *bp*2005 and *pl*2005 are constant variables within our cross-section, but have the same prefix as our value-varying variables.

```
reshape long bp pl, i(id) j(trialnum)
describe
list in 1/12, sepby(id)
```

```
(note: j = 1 2 3 4 5 2005)

Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      36
Number of variables                  14   ->       5
j variable (6 values)                     ->   trialnum
xij variables:
                     bp1 bp2 ... bp2005   ->   bp
                     pl1 pl2 ... pl2005   ->   pl
-----------------------------------------------------------------------------


Contains data
  obs:            36
 vars:             5
```

```
 size:           432
--------------------------------------------------------------------------------------------------
               storage   display    value
variable name    type    format     label      variable label
--------------------------------------------------------------------------------------------------
id               byte    %3.0f                 ID of person
trialnum         int     %9.0g
age              byte    %3.0f                 Age of person
bp               float   %9.0g
pl               float   %9.0g
--------------------------------------------------------------------------------------------------
Sorted by: id  trialnum
     Note: Dataset has changed since last saved.

     +---------------------------------+
     | id   trialnum   age    bp    pl |
     |---------------------------------|
  1. |  1          1    40   115    54 |
  2. |  1          2    40    86    87 |
  3. |  1          3    40   129    93 |
  4. |  1          4    40   105    81 |
  5. |  1          5    40   127    92 |
  6. |  1       2005    40   112    81 |
     |---------------------------------|
  7. |  2          1    30   123    92 |
  8. |  2          2    30   136    88 |
  9. |  2          3    30   107   125 |
 10. |  2          4    30   111    87 |
 11. |  2          5    30   120    58 |
 12. |  2       2005    30   119    90 |
     +---------------------------------+
```

We did not intend to reshape *bp*2005 and *pl*2005.

To rememdy, we need to specify the values within j() with j(trialnum 1-5) to indicate that we only want to reshape suffixes with 1-5.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
reshape long bp pl, i(id) j(trialnum 1-5)
describe
list in 1/10, sepby(id)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      30
Number of variables                  14   ->       7
j variable (5 values)                     ->   trialnum
xij variables:
                        bp1 bp2 ... bp5   ->   bp
                        pl1 pl2 ... pl5   ->   pl
-----------------------------------------------------------------------------


Contains data
  obs:            30
 vars:             7
 size:           450
----------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
----------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
trialnum        byte    %9.0g
age             byte    %3.0f                 Age of person
bp              int     %3.0f
bp2005          float   %9.0g
pl              int     %3.0f
pl2005          float   %9.0g
----------------------------------------------------------------------------------
Sorted by: id  trialnum
     Note: Dataset has changed since last saved.

     +---------------------------------------------------+
     | id   trialnum   age    bp   bp2005    pl   pl2005 |
     |---------------------------------------------------|
  1. |  1          1    40   115      112    54       81 |
  2. |  1          2    40    86      112    87       81 |
  3. |  1          3    40   129      112    93       81 |
  4. |  1          4    40   105      112    81       81 |
  5. |  1          5    40   127      112    92       81 |
     |---------------------------------------------------|
  6. |  2          1    30   123      119    92       90 |
  7. |  2          2    30   136      119    88       90 |
  8. |  2          3    30   107      119   125       90 |
  9. |  2          4    30   111      119    87       90 |
 10. |  2          5    30   120      119    58       90 |
```

```
+-----------------------------------------------------+
```

Problem 4: No id varible to identify the cross-sectional unit

If each wide-format observation is a unique cross-sectional unit, then we can use _n to generate a unique id variable.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_wide3, clear
describe
gen idcode = _n
describe
reshape long bp pl, i(id idcode) j(trialnum 1-5)
describe
list in 1/10, sepby(id)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell



Contains data from cardio_wide3.dta
  obs:             6
 vars:            14                          31 Dec 2009 15:46
 size:           168
----------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
----------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
age             byte    %3.0f                 Age of person
bp1             int     %3.0f                 Blood pressure systolic Trial 1
bp2             int     %3.0f                 Blood pressure systolic Trial 2
bp3             int     %3.0f                 Blood pressure systolic Trial 3
bp4             int     %3.0f                 Blood pressure systolic Trial 4
bp5             int     %3.0f                 Blood pressure systolic Trial 5
bp2005          float   %9.0g
pl1             int     %3.0f                 Pulse: Trial 1
pl2             byte    %3.0f                 Pulse: Trial 2
pl3             int     %3.0f                 Pulse: Trial 3
pl4             int     %3.0f                 Pulse: Trial 4
pl5             byte    %3.0f                 Pulse: Trial 5
pl2005          float   %9.0g
----------------------------------------------------------------------------------------------
Sorted by:
```

```
Contains data from cardio_wide3.dta
  obs:            6
 vars:           15                           31 Dec 2009 15:46
 size:          216
-------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
age             byte    %3.0f                 Age of person
bp1             int     %3.0f                 Blood pressure systolic Trial 1
bp2             int     %3.0f                 Blood pressure systolic Trial 2
bp3             int     %3.0f                 Blood pressure systolic Trial 3
bp4             int     %3.0f                 Blood pressure systolic Trial 4
bp5             int     %3.0f                 Blood pressure systolic Trial 5
bp2005          float   %9.0g
pl1             int     %3.0f                 Pulse: Trial 1
pl2             byte    %3.0f                 Pulse: Trial 2
pl3             int     %3.0f                 Pulse: Trial 3
pl4             int     %3.0f                 Pulse: Trial 4
pl5             byte    %3.0f                 Pulse: Trial 5
pl2005          float   %9.0g
idcode          double  %10.0g
-------------------------------------------------------------------------------------
Sorted by:
     Note: Dataset has changed since last saved.


Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        6   ->      30
Number of variables                  15   ->       8
j variable (5 values)                     ->   trialnum
xij variables:
                        bp1 bp2 ... bp5   ->   bp
                        pl1 pl2 ... pl5   ->   pl
-----------------------------------------------------------------------------


Contains data
  obs:           30
 vars:            8
 size:          690
```

```
-----------------------------------------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
-----------------------------------------------------------------------------------------------
id              byte    %3.0f                 ID of person
idcode          double  %10.0g
trialnum        byte    %9.0g
age             byte    %3.0f                 Age of person
bp              int     %3.0f
bp2005          float   %9.0g
pl              int     %3.0f
pl2005          float   %9.0g
-----------------------------------------------------------------------------------------------
Sorted by: id  idcode  trialnum
     Note: Dataset has changed since last saved.

     +-------------------------------------------------------------+
     | id   idcode   trialnum   age    bp   bp2005    pl   pl2005 |
     |-------------------------------------------------------------|
  1. |  1        1          1    40   115      112    54       81 |
  2. |  1        1          2    40    86      112    87       81 |
  3. |  1        1          3    40   129      112    93       81 |
  4. |  1        1          4    40   105      112    81       81 |
  5. |  1        1          5    40   127      112    92       81 |
     |-------------------------------------------------------------|
  6. |  2        2          1    30   123      119    92       90 |
  7. |  2        2          2    30   136      119    88       90 |
  8. |  2        2          3    30   107      119   125       90 |
  9. |  2        2          4    30   111      119    87       90 |
 10. |  2        2          5    30   120      119    58       90 |
     +-------------------------------------------------------------+
```

4.6 Multilevel datasets

Mitchell discusses when we have multiple levels of data, such as cross-sectional unit i and time dimension t, where data is level 1 data are constant across time within unit, but vary across cross-sectional units and level 2 data are time-varying data that vary within a cross-sectional units

Sometimes we have multiple levels, such as school district, school, classroom, and student.

If you are interested, please review pages 311-314

4.7 Collapsing datasets

The *collapse* command can be useful finding statistics at aggregate levels for different groups. You don't have to use egen but you can if you would like. After using the egen command, such as egen sumvar = sum(var), or egen avgvar = mean(var) the *collapse* command can be useful.

Let's say we want to aggregate mean wages by state in the CPS, we can use the *egen mean* command to find average wages in the state for month m in year y. We can then keep state month year and avgvar for all 50 states for each month and year of interest.

We need to be careful with the collapse command, since your original data are lost from memory unless you use a tempfile or a data frame.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
list, sepby(id)
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell

     +-----------------------------+
     | id   trial   age    bp    pl |
     |-----------------------------|
  1. |  1       1    40   115    54 |
  2. |  1       2    40    86    87 |
  3. |  1       3    40   129    93 |
  4. |  1       4    40   105    81 |
  5. |  1       5    40   127    92 |
     |-----------------------------|
  6. |  2       1    30   123    92 |
  7. |  2       2    30   136    88 |
  8. |  2       3    30   107   125 |
  9. |  2       4    30   111    87 |
 10. |  2       5    30   120    58 |
     |-----------------------------|
 11. |  3       1    16   124   105 |
 12. |  3       2    16   122    97 |
 13. |  3       3    16   101   128 |
 14. |  3       4    16   109    57 |
 15. |  3       5    16   112    68 |
     |-----------------------------|
 16. |  4       1    23   105    52 |
 17. |  4       2    23   115    79 |
 18. |  4       3    23   121    71 |
 19. |  4       4    23   129   106 |
```

```
 20. |  4       5    23   137    39 |
     |------------------------------|
 21. |  5       1    18   116    70 |
 22. |  5       2    18   128    64 |
 23. |  5       3    18   112    52 |
 24. |  5       4    18   125    68 |
 25. |  5       5    18   111    59 |
     |------------------------------|
 26. |  6       1    27   108    74 |
 27. |  6       2    27   126    78 |
 28. |  6       3    27   124    92 |
 29. |  6       4    27   131    99 |
 30. |  6       5    27   107    80 |
     +------------------------------+
```

Let's say we want to find mean blood pressure and pluse for each cross-sectional unit, then we can use the egen command or the collapse commands.

```
sort id
by id: egen meanbp = mean(bp)
by id: egen meanpl = mean(pl)
by id: egen maxbp = max(bp)
by id: egen maxpl = max(bp)
by id: egen minbp = min(bp)
by id: egen minpl = min(bp)
by id: egen medbp = pctile(bp), p(50)
by id: egen medpl = pctile(pl), p(50)

collapse meanbp meanpl maxbp maxpl minbp minpl medbp medpl, by(id)
```

Or you can just use the collapse command

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear

collapse (mean) meanbp=bp meanpl=pl (max) maxbp=bp maxpl=pl (min) minbp=bp minpl=pl (p50) medbp=b
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Notice that our data are rounded to the nearest 1, unlike our egen commands, we can format our data to reflect that decimal point to two places instead of rounding up by default

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use cardio_long, clear
format bp* pl* %5.2f
collapse (mean) meanbp=bp meanpl=pl (max) maxbp=bp maxpl=pl (min) minbp=bp minpl=pl (p5
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

4.8 Exercise

Pull BLS monthly inflation data, reshape it into long, and merge with CPS

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Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.